# It Only Takes *OneBite*: Attention Shocks in Local Markets*

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This version: November 5, 2026

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**Abstract:**

How does digital media redirect attention and influence consumer choice? In this paper, I investigate a novel setting and evaluate how online attention from a *OneBite* pizza review impacts consumer decisions and firm performance. Using variation in the review upload dates, I find that review scores $\geq$8.0 increase customers at the reviewed pizzeria by an average of 43% over the first three months following the review, while scores $< 8.0$ have no significant effect. Although a high-score review only directs attention to a single establishment, I show that the consumer response generates positive spatial spillovers onto nearby drinking places, consistent with consumption-based agglomeration from the shared customer base. Despite the initial increase, the attention does not create sustained growth as customers return to their pretreatment level within four months. To uncover the mechanisms that explain the consumer response, I develop a theoretical framework and decompose the response into three channels: awareness, information, and prestige. While the higher response from non-local consumers highlights the role of product awareness, I find consumers do not respond to new information about a product's quality. Instead, the prestige from visiting a viral establishment explains 83% of the immediate consumer response.

**Keywords: Attention Economy, Social Media, Spatial Spillovers**
**JEL Codes: D12, D83, M3, R12**

*Supported in part by funding from the Social Sciences and Humanities Research Council. I am incredibly grateful for the guidance of my committee, Nathaniel Baum-Snow, Robert Clark, Jonathan Hall, and Stephan Heblich, and for the helpful comments from Rawnak Amin, Dwayne Benjamin, Devin Bissky-Dziadyk, Alison Cane, Stephen Claassen, Victor Couture, Anthony McCanny, Fern Ramoutar, Noah Spencer, William Strange, Tianyi Wang, and many seminar participants. I also thank all participants from the 2026 CEA Meetings for comments and feedback. All errors are my own. Declaration of Interest: None.

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*In an information-rich world, the wealth of information means a dearth of something else: a scarcity of whatever it is that information consumes. What information consumes is rather obvious: it consumes the attention of its recipients. Hence a wealth of information creates a poverty of attention.*

— *Herbert A. Simon (1971)*

# 1 Introduction

Attention plays a prominent role in any decision (Serra-Garcia, 2026; Loewenstein and Wojtowicz, 2025; Bordalo et al., 2020). With seemingly infinite information in the modern economy, scarce attention governs consumer choice as it limits the information a person considers. This gives digital media the power to redirect consumer attention and shape market outcomes. Although understanding how attention influences choice is a fundamental concern for the attention economy, separately identifying the mechanisms that explain the response remains an empirical challenge.

In this paper, I use a novel setting to investigate how attention affects consumer decisions and separately identify three mechanisms: product information, product awareness, and media prestige. I focus on social media attention from a *OneBite* pizza review on YouTube, a video review in which Barstool Sports' Dave Portnoy samples a cheese pizza from a local pizzeria and scores the pizza from 0.0 to 10.0. Since the pizzeria does not anticipate the online review, the attention from the viral video review provides a quasi-exogenous demand shock to the firm. To evaluate the consumer response to the social media attention, I scrape all review scores and upload dates for pizzerias in the United States from both the *OneBite* app and YouTube and link the reviews to firm-level outcomes from SafeGraph's Spend Patterns dataset. Using a staggered event study framework, I estimate that total customers increase by 6.2% in the first month after the review, but immediately return to their pretreatment levels in the second month. The effects, however, critically depend on the review score. Following Portnoy's scoring guide, I estimate the heterogeneous treatment effects from the review and find that while total customers do not increase after a pizzeria receives a score $<$7.0 or in [7.0, 8.0), review scores $\geq$8.0 increase customers by 48%, 47%, and 34% in the first three months following the review. Despite the significant increase, the attention does not create sustained growth, as monthly customers return to their pretreatment level in the fourth month after the review.

I next evaluate the local market effects of a *OneBite* review and investigate whether the attention spills over onto nearby establishments, with a focus on the high-score re-

views. I first investigate whether consumers substitute away from local competitors and jointly estimate the direct treatment effect on the reviewed pizzeria with the spillover effects onto pizzerias in two treatment rings: within 500m and between 500m and 1,000m of the reviewed pizzeria. The specification estimates a treatment effect statistically similar to the main estimate and no detectable spillovers, showing limited evidence that consumers substitute away from other pizzerias. I next investigate whether the increase in customers generates positive spillovers for nearby establishments in other industries. Over the first three months following a high-score review, customers increase at nearby "Drinking Places (Alcoholic Beverages)" (+7%), consistent with consumption-based agglomeration as the local demand shock benefits establishments that offer complementary products and share a customer base with the reviewed pizzeria. However, customers decrease at "Limited-Service Restaurants" ($-7\%$), excluding other pizzerias.While I find no evidence that consumers substitute away from other pizzerias, the negative spillover suggests that customers substitute away from limited-service restaurants that serve other dining options.

Although the increase in customers shows that attention influences consumer decisions, prior knowledge of local alternatives may weaken a consumer's response to attention (Ackerberg, 2001). To investigate the role of market familiarity, I evaluate how the treatment effects vary with the distance from a consumer's home county to the pizzeria's county, a proxy for market familiarity. While high-score reviews generate increases in customers from all distance bins, the treatment effect is monotonic in distance, with the smallest effects for local consumers (+19%) who live in counties within 5 km and the largest increase from non-local customers (+66%) who live more than 50km from the pizzeria's county. The greater response from non-locals provides suggestive evidence that attention increases awareness of the reviewed pizzeria in the market.

Finally, I develop a theoretical framework to uncover the mechanisms that explain the consumer response. I adapt the product-specific bias model in Cerreia-Vioglio et al. (2023) to include both local consumers with equal awareness of all products and non-local consumers with unequal awareness across products. Within this framework, I identify three channels through which attention affects consumer choice: product information that captures updates to beliefs about product quality, product awareness that captures a consumer's increased consideration for reasons unrelated to utility, and media prestige that captures the consumption value an internet personality creates for consumers who visit a trending establishment. To separately identify the information channel, I exploit differences in the gap between the *OneBite* score and the pizzeria's Google rating and at-

tribute the remaining local consumer response to prestige. Under the assumption that only local consumers are equally aware of all products, the differential response of non-local consumers identifies changes in product awareness. Since the review score is always available online, I assume the information channel is persistent while prestige and awareness fade over time. Decomposing the reduced-form coefficients, I find that information does not significantly contribute to the consumer response. While awareness plays a larger role, the prestige from a high-score review explains 83% of the initial consumer response. The findings highlight how online personalities generate additional consumption value for products and shape consumer decisions. Although the prestige is substantial, attention does not generate long-term growth.

This paper makes three main contributions. First, I provide new causal evidence on how attention shocks affect local business performance and offer the first decomposition of the consumer response into three channels: information, awareness, and prestige. Understanding how attention affects firm performance remains a fundamental question in the literature as firms compete for scarce consumer attention Loewenstein and Wojtowicz (2025); Bordalo et al. (2016); Simon (1971). While the attention shocks increase consumer awareness about a product (Helmers et al., 2019; Honka et al., 2017; Clark et al., 2009; Goeree, 2008), *OneBite* reviews provide both information to consumers about product quality and a consumption value from visiting a pizzeria trending online. Related literature on advertising shows that information changes a consumer's perception of product quality (Dubé et al., 2026; Bagwell, 2007), and outlines the key distinction between prestige and information (Ackerberg, 2003, 2001). However, existing research does not separately identify the contributions of all three channels. My setting exploits unique variation in a review's timing and information update, and spatial variation in a consumer's location to separately identify how each channel contributes to the consumer response. The findings explain how a consumer responds to attention and how digital media shape local markets.

Second, I provide the first causal evidence that attention from online reviews generates positive demand spillovers onto nearby establishments. This connects the literature on online review platforms (Bao et al., 2024; Fang, 2022; Lawani et al., 2019; Kuang, 2017; Hu et al., 2008) with the literature on consumption-based agglomeration (Couture and Handbury, 2020; Couture, 2016; Glaeser et al., 2001). While the agglomeration literature often focuses on the productivity benefits from clustering, consumption-based agglomeration stems from a shared customer base. Specifically, a positive demand shock to one establishment attracts consumers to the area and increases demand at nearby establishments that offer complementary products. This paper provides evidence of the agglom-

eration benefits from online reviews, as nearby establishments benefit from the attention to a pizzeria that receives a high-score *OneBite* review.

Third, I provide new evidence on how media influences consumer decisions. Previous literature shows that radio (Wang, 2021) and television (Cornelson, 2018; Kearney and Levine, 2015; Ferrara et al., 2012) shape consumer behaviour through information. With social media's growing role as an information source (Allcott et al., 2022; Levy, 2021; Allcott and Gentzkow, 2017), the platforms create new opportunities for online personalities to influence consumer decisions, as audiences often interpret subjective media opinions as facts (Bursztyn et al., 2023). This paper investigates a novel setting where the online personality is an independent source of information and receives no payment for his review, which complements previous research on celebrity endorsements (Knoll and Matthes, 2017; Garthwaite, 2014; Elberse and Verleun, 2012; Agrawal and Kamakura, 1995). My theoretical framework separates the consumption value an online personality creates from changes in product awareness and expected quality, to highlight the significant prestige value from the media.

The remainder of the paper proceeds as follows. Section 2 provides an overview of *OneBite* reviews. Section 3 lays out a theoretical framework that microfounds the empirical specifications and highlights the mechanisms that underlie the treatment effect. Section 4 describes the data and sample. Section 5 outlines the empirical strategy and identification. Section 6 presents the main results. Section 7 investigates how each mechanism contributes to the consumer response. Finally, Section 8 concludes.

## 2 Context

Dave Portnoy, the founder and owner of Barstool Sports, started *OneBite* reviews in 2013 after a debate with friend and coworker Daniel Katz on what food item each would eat for the rest of their lives. To test his choice, Portnoy decided to eat only pizza for a month, film a review of each pizzeria, and upload each review to YouTube. In each video review, Portnoy eats "one bite" of cheese pizza directly outside the pizzeria and, following a brief discussion, rates the pizza from 0.0 to 10.0.[1] Under his scoring system, Portnoy states that "a 7.0 means good pizza; if you live near it, you go to it," while "anything 8.0 or above, you drive to it." Across over 2,100 reviews, the viral video reviews have accumulated over 1.2

[1] All pizza reviews feature a cheese pizza, a product standardization that limits menu heterogeneity, making scores comparable across establishments and ensuring the rating reflects the pizzeria rather than the specific item ordered.

billion views on YouTube as of September 2026, establishing Portnoy as "one of the most influential people in the world of food social media" (Draper and Nierenberg, 2023).

Social media occupies a significant share of a person's leisure time (Allcott et al., 2022) and fundamentally shapes one's behaviour and how users acquire information (Levy, 2021; Allcott and Gentzkow, 2017). The time users spend on social media can also influence consumer spending, as social media ads are now the most common source of product discovery for persons aged 16-34, exceeding both television ads and word of mouth (Kemp, 2025). While *OneBite* reviews are not an explicit advertisement, the finding reflects the potential for social media influencers to draw attention to products, share information with followers, and spur online trends. Although influencers span across industries, food and restaurant influencers are well-suited to investigate how attention impacts consumer decisions because the choice of where to dine is a frequent decision with a defined geographic market, making attention on social media tractable to measure.

Within this space, Dave Portnoy's *OneBite* pizza reviews are among the most prominent series on social media and provide the ideal setting to investigate how attention affects consumer decisions for five reasons. First, Barstool Sports often posts *OneBite* reviews on its social media pages to reach its established audience of over 66 million monthly unique users (The Economist, 2026). Second, financial incentives are unlikely to influence the review score because, unlike celebrities or social media influencers with brand deals, Portnoy is not paid to review or endorse the pizzeria. Third, the high frequency of reviews and varying scores generate substantial variation in both the timing of reviews and the magnitude of attention shocks across pizzerias. Fourth, while many social media influencers are based in a single location, *OneBite* reviews cover most major cities across the United States because Portnoy conducts the reviews while travelling for Barstool Sports. The widespread geographic coverage, combined with the high review count, provides sufficient within-county variation to control for local economic conditions. Finally, since Portnoy does not announce the pizzerias he plans to visit, establishments cannot anticipate the *OneBite* review, making the demand shock quasi-exogenous.

# 3 Theoretical Framework

In this section, I lay out a theoretical framework that both microfounds my empirical specification and describes the mechanisms through which attention affects consumer choice. I adapt the product-specific bias model in Cerreia-Vioglio et al. (2023) to include two con-

sumer types: local consumers with equal awareness of all products and non-local consumers with unequal awareness across products. The product-specific bias captures differences in product awareness, providing a continuous analogue to bounded consideration models that explicitly model a consumer's consideration set (Helmers et al., 2019; Honka et al., 2017; Goeree, 2008). Although all products remain available for a consumer under the product-specific bias, a low awareness weight can make choice probability sufficiently small to approximate a product's exclusion from a consideration set. Under the assumptions described below, the model identifies three potential channels through which attention affects consumer demand: an information channel from the quality signal, a prestige channel from the consumption value of visiting a trending pizzeria, and an awareness channel from the increased consideration of the reviewed pizzeria.

## 3.1 Basic Setup

Each month, a consumer $i$ in local market $c$ chooses from $J_{ct}$ pizzeria options and her outside option of dining at home or a non-pizzeria. Although true quality is unknown, a consumer maximizes her utility based on the quality signal from a pizzeria's average online rating $\widetilde{R}_j$, the time-invariant price $p_j$, and an i.i.d. Type I extreme-value preference shifter, $\varepsilon_{ijgct}$.[2] Consumers are either local $(L)$, with complete and equal awareness about all products in the market, or non-local consumers $(N)$, with unequal awareness about the products. Although product awareness differs, both consumer types $g \in \{L, N\}$ share a common utility function.

If the pizzeria receives a *OneBite* pizza review, a consumer's utility shifts through two channels: information and prestige. After seeing a *OneBite* review with score $s_j$, a consumer updates her expected quality relative to the average online rating $\widetilde{R}_j$. The change in expected quality captures the information from a *OneBite* review, with $\eta$ governing its effect on consumer utility. Since the *OneBite* review score is always available to consumers after the review posting date, $\tau_j$, I model the information channel as a persistent effect. In addition, I allow consumers to receive additional utility from visiting a trending pizzeria: the prestige channel, $\phi$. However, as social media trends may fade over time, I allow prestige to decay at rate $0 < \delta_p < 1$. With the two channels, a consumer $i$ of type $g \in \{L, N\}$ obtains the following utility from choosing pizzeria $j$ in market $c$ at time $t$:

[2]The stability assumption of the online ratings follows from Anderson and Magruder (2012), who find that within-restaurant ratings only drift by several hundredths of a star. To validate the stability assumption, I match individual Google reviews from Li et al. (2022) to my pizzeria sample and find that the within-pizzeria standard deviation of monthly Google review averages is 0.08 stars, consistent with treating $R_j$ as time-invariant. I assume prices are time-invariant within pizzerias, reflecting a data limitation.

$$u_{ijgct} = \underbrace{\widetilde{R}_j - \alpha p_j + \eta \mathbb{1}(t \geq \tau_j)(s_j - \widetilde{R}_j) + \sum_{\ell=0}^{\infty} \delta_p^{\ell} \phi \mathbb{1}(t = \tau_j + \ell)}_{V_{jct}} + \varepsilon_{ijgct}$$

For local consumers who are equally aware of all products in the market, the conditional choice probabilities follow from a canonical discrete choice framework, where $V_{jct}$ is the deterministic component of utility.[3]

$$\pi_{jLct} = \frac{\exp(V_{jct})}{1 + \sum_{j' \in J_{ct}} \exp(V_{j'ct})}$$

I scale the conditional choice probability by the number of local consumers in market $c$, $M_{Lct}$, to generate an expression for log demand from local consumers at pizzeria $j$ in Equation 1. The structure microfounds an event study specification and shows that local demand can be expressed as an establishment fixed effect for locals, county-date interacted fixed effects to absorb the local consumer market size and inclusive value, and the change in demand from a *OneBite* review. Under this structure, each event-time coefficient captures the change in customers after the review from the change in expected quality and the prestige of visiting a trending pizzeria.

$$\ln Q_{jLct} = \underbrace{\widetilde{R}_j - \alpha p_j}_{\mu_{jL}} + \underbrace{\ln M_{Lct} - IV_{Lct}}_{\Gamma_{L \times c(j) \times t}} + \sum_{\ell=0}^{\infty} (\eta(s_j - \widetilde{R}_j) + \delta_p^{\ell} \phi) \mathbb{1}(t = \tau_j + \ell) \tag{1}$$

## 3.2 Product-Specific Awareness for Non-Local Consumers

While local consumers have equal awareness of all products in the market, non-local consumers, $N$, are unequally aware and place an awareness bias $a_{jct}$ on each pizzeria. The bias captures relative awareness across alternatives and may reflect prior information, memory,

[3] In Cerreia-Vioglio et al. (2023) without product-specific bias, the choice probability is $\pi_{jLct} = \exp(V_{jct}/\lambda)/[1 + \sum_{j' \in J_{ct}} \exp(V_{j'ct}/\lambda)]$, where $\lambda$ is the unit cost of processing information. I hold $\lambda$ common across consumer groups and normalize it to one.

immediate perception, or advertising. A positive bias term does not imply that non-locals are more aware of that product than local consumers. Rather, it increases the choice probability relative to other products, for reasons unrelated to utility.[4] I model the awareness bias as a function of a firm-specific component, $\widetilde{a}_j$, and the shock from a *OneBite* review, $\kappa$. Similar to the prestige channel, I allow the awareness shock to decay over time, now at rate $0 < \delta_a < 1$.

$$a_{jct} = \widetilde{a}_j + \kappa \sum_{\ell=0}^{\infty} \delta_a^{\ell} \mathbb{1}(t = \tau_j + \ell)$$

With awareness bias, a non-local consumer's conditional choice probability for product $j$ is:

$$\pi_{jNct} = \frac{\exp\left(V_{jct} + a_{jct}\right)}{1 + \sum\limits_{j' \in J_{ct}} \exp\left(V_{j'ct} + a_{j'ct}\right)}$$

As with the log demand derivation for local consumers, I scale each product's conditional choice probability by the number of non-local consumers in the market $M_{Nct}$ to derive product $j$'s log demand from non-local consumers. Equation 2 presents the expression that follows a similar event-study framework to the local consumer log demand equation. However, the microfoundations for non-local consumers reveal a third channel through which a review affects total customers: the awareness channel that shifts the product-specific bias.

[4] Although lower awareness among non-local consumers may suggest negative biases and a higher probability of choosing the outside option, the interpretation of $a_{jct}$ depends on the choice environment. If visiting pizzeria $j$ requires commuting to the market, dining at home may be an attractive outside option, consistent with $a_{jct} < 0$. Instead, if the choice is conditional on already being in market $c$, returning home may be less attractive than for local consumers, making $a_{jct} > 0$ plausible. In this derivation, I normalize the outside-option bias to zero, so $a_{jct}$ captures relative rather than absolute awareness. However, the empirical specification does not require this normalization, as its contribution is absorbed into the fixed effects and does not influence the parameter estimates.

$$\ln Q_{jNct} = \underbrace{\widetilde{R}_j - \alpha p_j + \widetilde{a}_j}_{\mu_{jN}} + \underbrace{\ln M_{Nct} - IV_{Nct}}_{\Gamma_{N\times c(j)\times t}} + \sum_{\ell=0}^{\infty}\left[\eta(s_j - \widetilde{R}_j) + \delta_p^\ell \phi + \delta_a^\ell \kappa\right] \mathbb{1}(t = \tau_j + \ell) \tag{2}$$

Equations 1 and 2 describe demand from locals and non-locals, respectively. The treatment effect from an event study that evaluates how customers change after a *OneBite* review captures a composite of the three channels: the total customer response to information and prestige, and the increase in awareness for non-local consumers. Under the assumption that local consumers have equal awareness of products, the awareness term captures the response of non-locals to increased product consideration. If both local and non-local consumers had unequal awareness across products, the difference in the parameters would capture the differential response to the change in product awareness.

### 3.3 Mechanisms

Although the microfoundations for each consumer type's log demand show that an event study captures the aggregate consumer response, the aggregate response does not separately identify each mechanism. To identify each channel, I stack Equations 1 and 2 to create a triple-difference specification, where three differences arise from reviewed versus unreviewed pizzerias, post-review versus pre-review periods, and local versus non-local responses. Equation 3 presents the specification and highlights the three channels.

$$\begin{aligned}\ln Q_{jgct} = &\underbrace{\widetilde{R}_j - \alpha p_j + \mathbb{1}(g = N)\widetilde{a}_j}_{\mu_{jg}} + \underbrace{\ln M_{gct} - IV_{gct}}_{\Gamma_{g\times c(j)\times t}} \\ &+ \underbrace{\eta(s_j - \widetilde{R}_j)}_{\text{Information}} \mathbb{1}(t \geq \tau_j) \\ &+ \sum_{\ell=0}^{\infty} \underbrace{\delta_p^\ell \phi}_{\text{Prestige}} \mathbb{1}(t = \tau_j + \ell) \\ &+ \sum_{\ell=0}^{\infty} \underbrace{\delta_a^\ell \kappa}_{\text{Awareness}} \mathbb{1}(g = N)\mathbb{1}(t = \tau_j + \ell)\end{aligned} \tag{3}$$

Equation 3 shows that the demand equation for each group is governed by consumer type-establishment and consumer type-market-time fixed effects, and three treat-

ment channels. The first, $\eta(s_j - \widetilde{R}_j)$, captures the demand response from new information about the pizzeria's expected quality. The second, $\delta_p^\ell \phi$, captures the response from the prestige of visiting a trending pizzeria. Finally, $\delta_a^\ell \kappa$ captures the additional demand response from increased awareness among non-local consumers. The theoretical framework separately identifies each channel using different sources of variation.

Since the information channel is persistent, the level shift in the treatment effects across time reflects the value of the information. Furthermore, the information response varies across pizzerias because the information update is larger when the score is further from the pizzeria's online rating. For instance, scoring 8.0 on a *OneBite* review generates a larger information update to a restaurant with a 1-star rating on Google, compared to a restaurant with a 3-star rating.[5] Both the persistence in the treatment effect and the variation in score updating across establishments allow for $\eta$ to be separately identified from both awareness and prestige shocks. After accounting for the information update, the prestige channel captures the remaining demand response from local consumers. Since prestige decays over time, the initial shock, $\phi$, is identified based on the remaining response in the initial treatment month, while the change over time captures the decay rate.

The awareness and awareness decay parameters are separately identified from the prestige channel based on the differential response from non-local consumers. Equation 4 shows differenced log demand equations for local and non-local consumers from Equations 1 and 2 to show how the triple-difference equation identifies the awareness channel. Since the information and prestige channels enter the utility functions for each group in the same manner, and the fixed effects capture group-specific establishment baselines and market-time conditions, the remaining differential response between the consumer types comes from the shift in the non-local consumer's product bias. While the change in the treatment effect for non-locals relative to locals reflects the shift in product bias and awareness, the differential decay between the two consumer types identifies awareness decay.

$$\begin{aligned} \ln Q_{jNct} - \ln Q_{jLct} = (\mu_{jN} - \mu_{jL}) + (\Gamma_{N\times c(j)\times t} - \Gamma_{L\times c(j)\times t}) \\ + \sum_{\ell=0}^{\infty} \underbrace{\delta_a^\ell \kappa}_{\text{Awareness}} \mathbb{1}(t = \tau_j + \ell) \end{aligned} \tag{4}$$

[5] In the data, Google ratings and *OneBite* scores use different scales. I therefore rescale the Google rating to match the mean and variance of Portnoy's scores, $s_j$, to generate the information update. After this transformation, an 8.0 equates to a 4.61 Google rating.

Separately identifying the prestige channel from awareness requires the equal awareness assumption for local consumers. If both local and non-local consumers have unequal awareness, the awareness shock captures the differential response in product consideration between the two types. However, if both types experience an awareness shock, the change in demand from increased awareness cannot be separately identified from prestige, as both channels generate demand responses identified from the same variation in review timing. Under this assumption, the spatial variation in consumer home locations allows a triple-difference specification to separately identify three mechanisms that underlie the aggregate demand response.

# 4 Data

## 4.1 *OneBite* Review Data

To evaluate how attention online affects consumer decisions and firm outcomes, I construct a comprehensive dataset of all *OneBite* reviews. First, I scrape Portnoy's reviews from the *OneBite* app, launched in 2018 to consolidate his reviews and further engage with social media followers, and collect each reviewed pizzeria's name, address, score, and the review date.[6] One feature of the *OneBite* app is that it allows social media followers to submit scores and videos of any local pizzeria, including pizzerias Portnoy has not reviewed. Using the universe of app user reviews, I scrape the names and addresses of the pizzerias Portnoy did not review to construct a natural control group.[7] Although the *OneBite* app identifies treated pizzerias, their scores, and their treatment dates, the app contains only 1,319 reviews, fewer than the review count on the *OneBite* YouTube channel.

To complete the treatment sample, I scrape all videos on the *OneBite* YouTube channel from its launch through August 2025.[8] While each video's YouTube page includes information on the pizzeria's name, city, and state, the source code does not list the pizzeria's address or the review score. Instead, I use the video's infographic to manually record the pizzeria's address, city, and state, and identify its geographic coordinates using the

[6] Figure B1 presents an example of the mobile app's interface. The first image presents the map feature to help users locate nearby pizzerias and identifies pizzerias with *OneBite* reviews and other unreviewed pizzerias in the area. In contrast, the second image shows a sample pizzeria's page, which includes information such as its name, address, its *OneBite* review score, its app user score, and a link to the video review.

[7] Figure A1 presents kernel density plots for both average Google Rating and the number of Google reviews. The density plots show that while Portnoy selects pizzerias with higher average Google Ratings, the distribution of how prominent each pizzeria is in terms of reviews is similar across the two samples.

[8] Although *OneBite* review videos are subsequently shared on other social media platforms, the reviews first appear on YouTube.

Google Places API. To identify each pizzeria's review score, I submit each video's transcript to Anthropic's Claude API. The prompt instructs the model to identify Portnoy's score from common surrounding phrases, such as "I'm gonna go [score]" or "[score], that's a review," within otherwise unstructured text.[9] I then match the YouTube review data to the *OneBite* app using each pizzeria's name, location, and review date to complete the treatment sample. To validate the imputed scores from Claude, I compare the app score with the extracted score and find that the review scores are within 0.1 points for over 93% of matched pizzerias.[10] Although I extract the review score for every video, I use the *OneBite* app score when available and use the extracted transcript score only for pizzerias that do not appear on the app.

I supplement *OneBite* review data with Google ratings and review counts for each pizzeria to capture the observable quality signal potential customers face. Although the data is collected at a single point in time through the Google Places API, I obtain individual Google reviews posted through August 31, 2021, from Li et al. (2022)'s dataset to understand how individual users respond to *OneBite* reviews, and use the data to reconstruct the pre-treatment average Google rating for pizzerias reviewed before August 31, 2021.[11] Finally, I collect cumulative YouTube view counts for each review video to understand each review's reach on social media.[12] Although a complete view-count history is unavailable, I combine the cumulative counts with historical counts recovered from each video's archived pages in the Internet Archive's Wayback Machine, with an average of 5.1 captures per video review, to evaluate how attention changes over time.

Table 1 presents annual summary statistics for all *OneBite* reviews of pizzerias in the United States between January 2019 and August 2025. During the sample period, Portnoy reviewed 1,653 pizzerias, an average of 248 pizzeria reviews per year, with a median review score of 7.4. Interestingly, total review volume did not substantially decline during the COVID-19 pandemic, as Portnoy posted 232 pizzeria reviews in 2020, com-

[9] Appendix B.1 presents a full description of the prompt and model version for replicability purposes.

[10] Figure B3 presents a heatmap to show Claude's extracted score from a video's YouTube transcript compared to the app score for all matched pizzerias.

[11] For all pizzerias treated after August 31, 2021, I first attempt to extract average ratings from the Internet Archive. However, if pizzerias do not have an archived snapshot, I use the rating collected through the API. While this is a limitation if a *OneBite* review influences the average Google rating, Figure A2 presents a stacked event study to show how the observed online rating changes after a *OneBite* review using the individual-review data. The figureAlthough information from a video takes longer than reading a conventiona shows that six months after a review, the average rating changes by only $0.009$ stars, likely unobservable to a consumer since Google only displays ratings in $0.1$-star increments unless it is at a crossing threshold. The slow drift is mechanical, as each review receives a weight of only $1/N$ in the average Google rating.

[12] The cumulative view counts for each video were scraped on March 24, 2026.

pared with 240 in 2019. Although Portnoy supplemented the pizzeria video reviews with frozen pizza reviews during the temporary restaurant closures in 2020, many pizzerias quickly reopened for takeout options, which allowed regular pizzeria reviews to continue.[13] Throughout the sample period, the median score remains between 7.2 and 7.6, while the standard deviation marginally decreases in later years. Despite the decline in review score variance, high-score reviews remain consistently infrequent over time, with only 14.2% of pizzerias achieving a score $\geq$8.0 during the sample period.[14]

The final three columns in Table 1 present descriptive statistics for the YouTube video reviews. With an average length of 261 seconds, the video reviews likely take longer to watch than it takes a potential consumer to look up an online review score. However, the video review often contains more detailed information about the establishment and the product than a typical online review. Furthermore, Portnoy's outspoken and polarizing persona gives the video reviews entertainment value and the feel of reality television (Draper and Nierenberg, 2023). Although acquiring information from a video takes longer than reading a conventional review, the video's entertainment value may offset the cognitive cost of acquiring the information. Consistent with their entertainment value, the video reviews attract substantial viewership, with the average video amassing 312,000 views and the most viewed video accumulating nearly 4.7 million views as of March 24, 2026.[15] Alongside the information from the review scores, *OneBite* reviews increase local business exposure on social media, with viral content to spread product awareness and information to potential consumers.

To assess whether the reviews generate online attention, I collect weekly Google Trends search intensity data for reviewed pizzerias over a two-year window, centred on the review date.[16] The Google Trends search intensity index ranges from 0 to 100, with 100 denoting the maximum search intensity for the restaurant in the market area over the observed period. I normalize the index into a $z$-score to express changes in search intensity in standard deviation units, and use a stacked event study to estimate how search

[13] The descriptive statistics in Table 1 exclude the frozen pizza reviews that Portnoy conducted during the COVID-19 pandemic when restaurants could not remain open, and exclude any reviews on non-pizza products at the same establishments.

[14] Figure A3 presents box plots of *OneBite* review scores by year of upload from 2019 to 2025. The figure shows that while the median score remained similar over the sample period, the interquartile range decreased, consistent with the decreasing variance shown in Table 1.

[15] In Table 1, the view counts are measured as cumulative counts at a specific point in time. As such, videos in later years will list fewer views by construction.

[16] Although I attempt to extract data from the Google Trends API for all reviewed pizzerias, the API only returns search intensity data for 60.1% of reviewed pizzerias.

Table 1: Summary Statistics of *OneBite* Reviews

| | | Review Scores | | | Video Characteristics | | |
|---|---|---|---|---|---|---|---|
| Year | Reviews | Median | St.Dev | Share $\geq 8.0$ | Average Views (000s) | Max Views (000s) | Average Duration (sec) |
| 2019 | 240 | 7.2 | 1.40 | 15.1% | 386.3 | 4,694.0 | 286.9 |
| 2020 | 232 | 7.3 | 1.58 | 12.9% | 349.8 | 3,149.2 | 314.8 |
| 2021 | 243 | 7.3 | 0.94 | 11.4% | 288.7 | 2,226.1 | 249.6 |
| 2022 | 250 | 7.4 | 1.11 | 15.7% | 337.8 | 2,282.6 | 247.2 |
| 2023 | 248 | 7.4 | 1.04 | 13.0% | 341.2 | 3,779.0 | 249.3 |
| 2024 | 236 | 7.6 | 0.59 | 16.5% | 254.8 | 949.9 | 245.6 |
| 2025 | 204 | 7.4 | 0.85 | 14.6% | 207.2 | 1,656.5 | 230.6 |
| **Total** | **1,653** | **7.4** | **1.13** | **14.2%** | **311.9** | **4,694.0** | **260.8** |

Notes: Table 1 presents the annual summary statistics for *OneBite* video reviews between 2019 and 2025. While the first column lists the total number of *OneBite* reviews posted to YouTube in each year, the next 3 columns present the annual summary statistics for the review scores only for reviews where the score is labelled either from the *OneBite* app or extracted from the YouTube Transcript. Figure A16 presents a stacked bar chart showing the annual number of reviews by score category to complement this table. The final 3 columns summarize video characteristics, including the average number of views and maximum number of views for videos in each year as of March 24, 2026, and the average duration of each video. Since views are calculated from a point in time, more recent years tend to list lower view counts on the videos by construction.

intensity for a pizzeria changes within its designated market area after a *OneBite* review is uploaded to YouTube.[17] Figure 1 presents the results and shows that search intensity for the reviewed pizzeria increases by 1.2 standard deviations within the pizzeria's designated market area in the first week after a *OneBite* review.[18] Despite the initial shock, the surge in search traffic is transitory and online searches return to their pre-review levels within six weeks. However, the increase in search intensity demonstrates that a *OneBite* review generates significant media attention towards the reviewed pizzerias.[19]

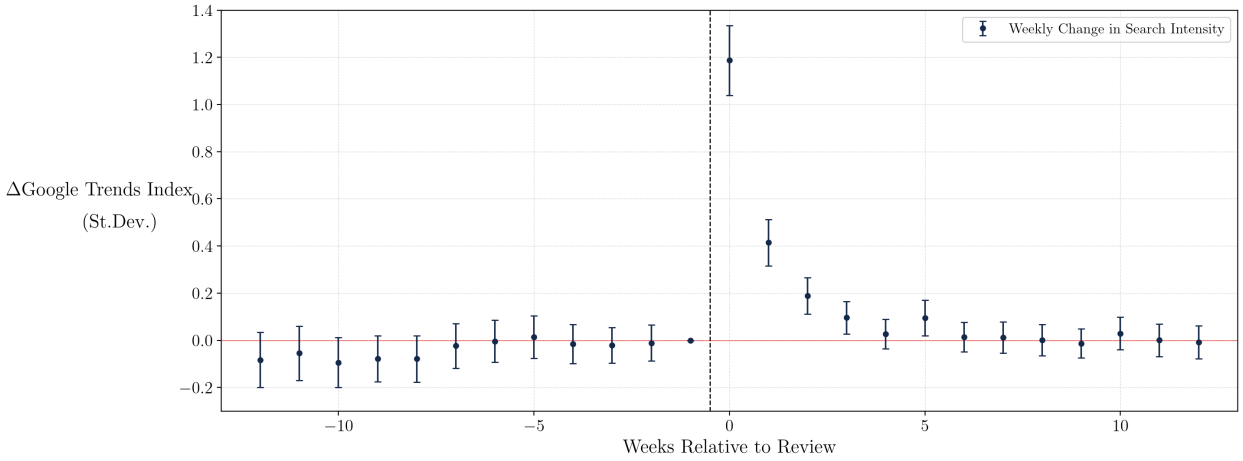



Figure 1: Local Google Search Trends

Notes: Figure 1 presents stacked event study estimates of the dynamic effect of a *OneBite* review on local search attention, measured using Google Trends data within a designated market area. The estimates show the weekly change in Google Trends search index in standard deviations in the 13 weeks before and after a review is posted, estimated on the geographic Google Trends data from 1263 reviews across 95 Designated Media Markets (DMAs). Reviews are excluded from the sample when the pizzeria's location cannot be assigned to a DMA or when Google Trends returns no DMA-level search data for the query window. The search index ranges from 0 to 100 across DMAs within each time-window query. The week prior to the review is normalized to 0. The confidence bars show the 95% confidence intervals with standard errors clustered at the DMA level.

[17]The designated market area (DMA) classifications used by Google follow Nielsen's definitions. DMAs group counties together based on which local television stations account for a dominant share of viewing. DMAs are the standard geographic definition used for targeted advertising by geographic regions, local audience measurements, and measuring market-specific data on viewing habits and preferences.

[18]Figure A17 presents an analogous event study of Google Trends responses across Portnoy's three score categories. Figure A18 presents a difference-in-discontinuities regression to examine the Google Trends response around the 8.0 high-score cutoff.

[19]Figure A4 presents an alternative measurement that uses a stacked PPML event study to estimate how the total number of posted Google reviews changes after a *OneBite* review is uploaded to YouTube, using Google review data from Li et al. (2022). Similarly, Figure A5 presents the results for how the number of Google reviews that contain the words "Barstool Sports", "Portnoy", or "*OneBite*" changes after a review is uploaded. However, given the infrequency of the use of the words in the pretreatment period, the treatment effects should be interpreted with caution.

## 4.2 SafeGraph's Spend Patterns Data

To investigate how online attention affects firm performance, I acquire monthly firm-level transaction data from SafeGraph's Spend Patterns dataset for January 2019 through August 2025. The dataset includes anonymized transaction data for points of interest, with firm-level monthly data on total customers, total spending, and total transactions. Although reported at the firm level, the dataset is derived from a consumer panel covering 4% of U.S. consumers and captures only a share of each establishment's economic activity. However, given the representative consumer sample, this scales observed activity proportionally across establishments. Furthermore, since the data are based on an underlying consumer panel, the dataset provides customer counts by city of origin. I supplement the Spend Patterns dataset with SafeGraph's Global Places dataset to identify each establishment's name, address, coordinates, and North American Industry Classification System (NAICS) code. Since the total number of captured customers fluctuates based on changes in the financial institutions that provide data to SafeGraph, I follow SafeGraph's normalization procedure and reweight each observation by the average number of consumers in each state $s$ during the sample period divided by the total number of consumers in the state in month $t$, or $\bar{N}_s/N_{st}$. The adjustment ensures that any change in customers, transactions, or spending represents a change in firm performance rather than a change in sample size.

Although *OneBite* video reviews cover pizzerias in 39 U.S. states and in Canada, England, and Italy, I restrict the sample to the 20 states with more than 15 video reviews, or more than 10 reviews since 2019. The state restriction ensures a sufficiently large sample of establishments to control for local economic trends after I match to the SafeGraph sample. Table A1 reports the total review counts for each state in the restricted sample. Since Portnoy often reviews pizzerias when he travels for Barstool Sports, the reviewed pizzerias are clustered in states with large metropolitan areas, including New York (666), New Jersey (221), Florida (194), and Massachusetts (161).

With the state-restricted sample, I match the *OneBite* reviews to the SafeGraph spend patterns dataset using each establishment's name and address to construct a pizzeria-month panel. Since streets often have multiple name designations (for example, New Jersey Turnpike and I-95), I then match any remaining establishments based on whether a reviewed pizzeria's geographic coordinates are within 20m of a SafeGraph establishment with the same name, a narrow radius to minimize false matches. The two-stage procedure matches 744 pizzerias that receive *OneBite* pizza reviews to SafeGraph. However, I omit

Minnesota in the final sample because I do not match any reviewed pizzeria to an establishment in SafeGraph's Spend Patterns dataset.[20] Finally, I use the Google Places API to identify whether each establishment is a restaurant and to exclude any pizza reviews from hotels, grocery stores, or other establishments because firm-level outcomes cannot isolate changes in demand for the restaurant from other components of the business. After this restriction, the final treatment sample includes 679 reviewed pizzerias. I apply the same procedure to match user-reviewed pizzerias on the *OneBite* app to SafeGraph and create a control sample of 5,689 pizzerias.[21] Figure 2 maps the matched pizzerias in New York City. Although fan-reviewed pizzerias are more dispersed relative to the treated pizzerias, the geographic dispersion provides every treated pizzeria with nearby control establishments.

Although the 679 treated pizzerias provide a sufficiently large sample for empirical analysis, the sample restriction excludes 67% of all video reviews. Figure A6 presents kernel density plots of scores reported on the *OneBite* app, scores extracted from YouTube transcripts using Claude, and scores in the final matched sample to assess whether the matched pizzerias are representative of the full set of *OneBite* video reviews. The figure presents two results. First, the kernel density plots for app scores and Claude-extracted YouTube scores share an identical median of 7.4 and comparable standard deviations, which suggests that Claude recovers a similar score distribution to the app. Second, after restricting to the matched sample, the review score distribution remains near-normal with an identical median and a similar, though slightly smaller, standard deviation. The similarity across the distributions suggests that selection into the matched sample is unrelated to review score and is consistent with incomplete SafeGraph coverage rather than score-based selection.

# 5 Empirical Strategy

## 5.1 Staggered Event Study

Motivated by the theoretical framework, I exploit the quasi-exogenous variation in each *OneBite* review's upload date and estimate a staggered event study to evaluate how attention influences consumer choices. Equation 5 presents the reduced-form specification,

[20] Table B1 reports summary statistics for observed customer counts and total spending in the final sample, which reflects activity captured by SafeGraph's consumer sample rather than establishment totals.

[21] One limitation is that SafeGraph's privacy agreements with data providers suppress some establishment-month records, causing sample entry and exit unrelated to economic activity. Section B.2 examines whether review-induced changes in sample composition could explain the treatment effects and finds no evidence that reviews affect sample entry or exit.

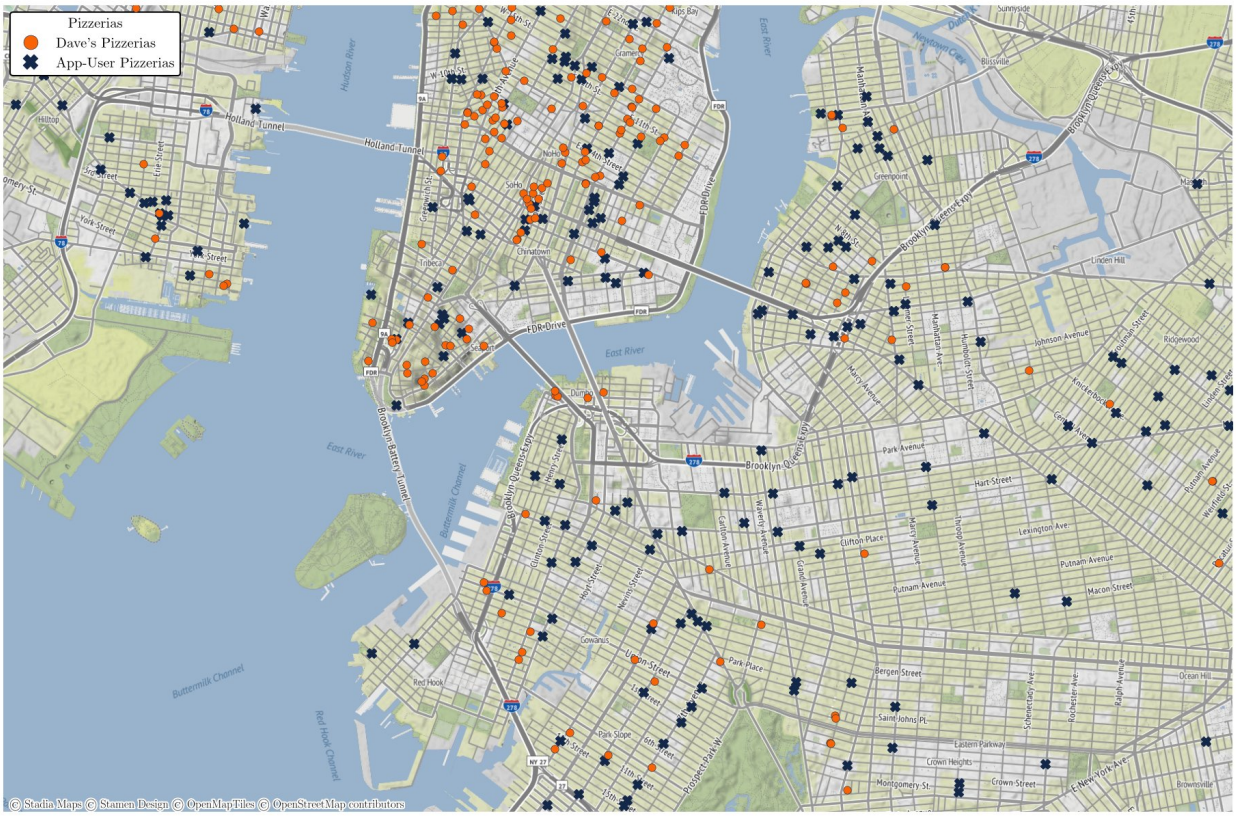



Figure 2: *OneBite* Pizza Reviews around New York City

Notes: Figure 2 presents the spatial distribution of *OneBite* reviews in New York City. Dave's pizzerias show the location of all pizzerias with a *OneBite* review, while app-user-reviewed pizzerias serve as the untreated control group. The sample includes all reviews posted between 2013 and 2025. Pizzeria coordinates are geocoded from the address in the *OneBite* app when available, and from the video infographic on YouTube otherwise, using the Google Maps Geocoding API.

where $Q_{jct}$ is the total number of customers at pizzeria $j$ in month $t$, $\mathbb{1}(t = \tau_j + \ell)$ is an event-time indicator equal to one when $t$ is $\ell$ months from the *OneBite* review, $\tau_j$ is the month when the review is posted, $\mu_j$ is a pizzeria fixed effect, $\mu_t$ is a month fixed effect, and $\Gamma_{c(j)\times t}$ is a county-month fixed effect, where pizzeria $j$ is located in county $c(j)$.[22] While $\tau_j$ denotes the month when the review is posted, review dates are at the daily level. Following the theoretical framework, I use total monthly customers as the firm performance metric to measure the unique consumers who visit a pizzeria in a given month and capture how many customers the attention draws to the establishment. However, I also estimate how a *OneBite* review impacts total monthly spending at the pizzeria to investigate the effects on firm revenues and report the results in Appendix A. Since total customers is a non-negative count variable, I estimate Equation 5 using Poisson pseudo-maximum likelihood (PPML) to prevent the bias from log-linear transformations under heteroskedasticity (Santos Silva and Tenreyro, 2006).

$$Q_{jct} = \exp\left(\mu_j + \mu_t + \Gamma_{c(j)\times t} + \sum_{\substack{\ell=-5 \\ \ell\neq -1}}^{5} \beta_\ell \mathbb{1}(t = \tau_j + \ell)\right) + \varepsilon_{jct}. \tag{5}$$

The coefficients of interest, $\{\beta_\ell\}_{\ell=0}^{5}$, estimate the change in total customers $\ell$ months from the review relative to $\ell = -1$.[23] I include the county-month fixed effects to control for unobserved and heterogeneous variation in pizzeria customers at a local level. Local variation is a potential confounder in the restaurant industry during the sample period because local governments both prohibited and reopened in-person dining on different dates in response to the COVID-19 pandemic. The interacted fixed effects absorb the policy differences in restaurant activity between counties.[24] Finally, I restrict the event study window to five months on either side of the treatment date. While the Google Trends event study in Figure 1 shows an immediate increase in search intensity that reverted to pre-treatment levels within six weeks, the five-month post-treatment horizon therefore extends beyond the attention shock to capture any effects that persist after online attention fades.

[22] While I list both the county-by-month fixed effects $\Gamma_{c(j)\times t}$ and the month fixed effects $\mu_t$ in the written specification, $\Gamma_{c(j)\times t}$ are perfectly collinear with the month fixed effects $\mu_t$, so the two sets of fixed effects cannot be separately estimated.

[23] With a PPML specification, $100 * [\exp(\beta_\ell) - 1]$ gives the percentage change in total customers.

[24] Using a more microgeographic interaction such as the tract-level variation increases any bias in the treatment effect if consumers substitute away from other pizzerias in the local area. An estimating equation that exploits any within-tract variation would amplify bias more than a within-county equation. I discuss the stable unit treatment value assumption in Section 5.2.

To investigate how the treatment effect varies with the review score, I use the PPML specification in Equation 6 to jointly estimate all event study coefficients for each of Portnoy's three score categories: $\sigma \in \{H, M, L\}$. I follow Portnoy's score categorizations and define a high score $(H)$ as any review score of at least 8.0, a middle score $(M)$ as any score weakly larger than 7.0 but less than 8.0, and a low score $(L)$ as any score below 7.0.[25] In Equation 6, each $\beta_\ell^\sigma$ estimates how a score in $\sigma$ impacts firm performance $\ell$ months after the review.

$$Q_{jct} = \exp\left(\mu_j + \mu_t + \Gamma_{c(j)\times t} + \sum_{\sigma\in\{H,M,L\}} \sum_{\substack{\ell=-5 \\ \ell\neq -1}}^{5} \beta_\ell^\sigma \mathbb{1}(t - \tau_j = \ell, s_j = \sigma)\right) + \varepsilon_{jct}. \tag{6}$$

The effect of a *OneBite* review may vary for three reasons, each captured by a channel in the theoretical framework. First, reviews with different scores may generate different amounts of awareness. Among the ten most viewed *OneBite* video reviews, five reviewed pizzerias received scores below $7.0$ and five received scores greater than or equal to $8.0$, while none earned a middle-score review. The concentration in the outer score bins indicates the relationship between score and online attention is non-monotonic, with significant attention from both highly positive and highly negative reviews. Second, the review score may update a consumer's belief about a pizzeria's quality. Although the scores do not provide an objective measure of quality, consumers often interpret subjective opinions as objective facts (Bursztyn et al., 2023). For instance, a high *OneBite* score may generate a larger increase in customers than a low score because it provides a more positive update relative to a consumer's prior beliefs about the pizzeria. Finally, the consumption value of visiting the pizzeria may differ across score groups, and consumers may only derive utility from a visit to a pizzeria that Portnoy awards a high score.

## 5.2 Identification

Since a *OneBite* review's timing is staggered, I use a stacked event study design following Cengiz et al. (2019) to allow for the treatment effects to vary across cohorts and time. The conventional two-way fixed-effects estimator will not recover the average causal effect when treatment effects vary across treatment cohorts because the variation that identifies the two-way fixed-effects coefficients uses already-treated units as controls for later-

[25] Formally, the score categories can be written as $\sigma \in \{[8.0, 10.0], [7.0, 8.0), [0.0, 7.0)\}$.

treated units (Roth et al., 2023; Goodman-Bacon, 2021). Although numerous methods exist to allow for heterogeneous treatment effects that vary across cohorts and ensure already-treated units do not enter the control sample (Borusyak et al., 2023; Gardner, 2022; Sun and Abraham, 2021; Callaway and Sant'Anna, 2021; Wooldridge, 2021), a few corrections accommodate non-linear specifications such as PPML.[26] I construct the stacked panel as follows.

For each treatment month, I construct a subsample that contains all pizzerias that receive a *OneBite* review during that month and control pizzerias that remain untreated through the end of the estimation window. The restriction excludes already treated pizzerias and any not-yet-treated establishments that receive a review within six months of the current cohort's treatment date to prevent bias from using treated pizzerias as controls.[27] After restricting each subsample to its estimation window, five months before and five months after the treatment date, I stack the subsamples into a single dataset. Following Cengiz et al. (2019), I interact all fixed-effect variables in my specification with a stack-specific ID variable to identify the treatment effects using the variation within each treatment cohort. Finally, I cluster standard errors at the pizzeria level to allow for observations at the same pizzeria to be correlated across subsamples. Using the stacked dataset, I estimate the PPML specification following Correia et al. (2020), which estimates the treatment effects of a *OneBite* review without the bias from using already treated units as controls, and without heteroskedasticity concerns from using non-negative count data. While I use the stacked PPML as the main specification, I estimate log-linear specifications with corrections for time and cohort treatment effect heterogeneity following a stacked linear event study, Borusyak et al. (2023), Gardner (2022), and Sun and Abraham (2021) as robustness checks.

Although Portnoy does not review pizzerias at random, the staggered event study

[26] For instance, both Borusyak et al. (2023) and Gardner (2022) construct counterfactual outcomes using a two-stage imputation procedure. The first stage estimates all fixed effects in the specification using all untreated observations in the sample. While including never-treated observations provides variation to identify all time fixed effects in the sample, the not-yet-treated observations provide the relevant variation to identify all individual fixed effects without any confounding factors from treatment. The estimated fixed effects from the first stage are then used to either impute the counterfactual outcome directly (Borusyak et al., 2023), or residualize the outcome variable before the second stage (Gardner, 2022). However, the procedure cannot extend directly to PPML because the first-stage fitted values are biased in finite samples; bias from the fixed effects in the first stage carries through to the imputed treatment effects.

[27] Within each stack, I ensure control pizzerias lie in the common support across observables discussed in Baker et al. (2026). The restriction also limits the comparison group to observationally similar restaurants, mitigating concerns that restaurants Portnoy did not visit differ systematically in scale or price point from those he reviewed. As a robustness check, I include an additional specification with the full set of not-yet-treated observations in Figure A7.

identifies the causal effect of a *OneBite* review under three assumptions. First, I require that firm performance evolves similarly for reviewed and non-reviewed pizzerias before treatment: the parallel trends assumption. With a PPML specification, parallel trends assumes treated and untreated pizzerias follow the same proportional change in customers leading up to treatment. To evaluate the sensitivity of the estimates to violations of parallel trends, I follow Rambachan and Roth (2023) and provide alternative treatment effect bounds that remain valid under violations no larger than the observed pre-trends. Second, I require that each review's upload date is exogenous. The primary concern is that Portnoy systematically selects pizzerias that are currently experiencing a sudden increase in demand as he reviews them. Although the endogenous timing is possible, the steady pretrends in Google Trends in Figure 1 provide evidence that pizzerias on average do not experience any sudden increase in attention before a *OneBite* review. Third, I require the stable unit treatment value assumption (SUTVA), such that a review does not contaminate the outcomes of the control sample. If spatial spillovers from a *OneBite* review cause customers to substitute away from nearby pizzerias, the specification will overestimate the causal effect as control pizzerias become worse off after nearby restaurants are treated. Conversely, if a treated pizzeria faces a supply constraint and the review brings more customers to the area, customers may substitute toward nearby pizzerias; the specification will underestimate the causal effect. As a robustness check, I re-estimate Equation 5 after excluding pizzerias within 500m or 1,000m of a treated pizzeria.

# 6 Results

## 6.1 Consumer Response

Figure 3 presents the event study estimates to show how a *OneBite* review affects total monthly customers.[28] While the left vertical axis shows the treatment effect estimate, the right vertical axis converts the point estimate to a percentage change. Figure 3a shows that the online attention increases total customers at the pizzeria by 6.2% in the first month following the review. However, the increase is transitory, and the treatment effects become statistically similar to the pretreatment level in the second month following the review. The transitory increase aligns with the effects on Google Search intensity shown in Figure 1, which showed an instantaneous increase in online attention. While the dynamic

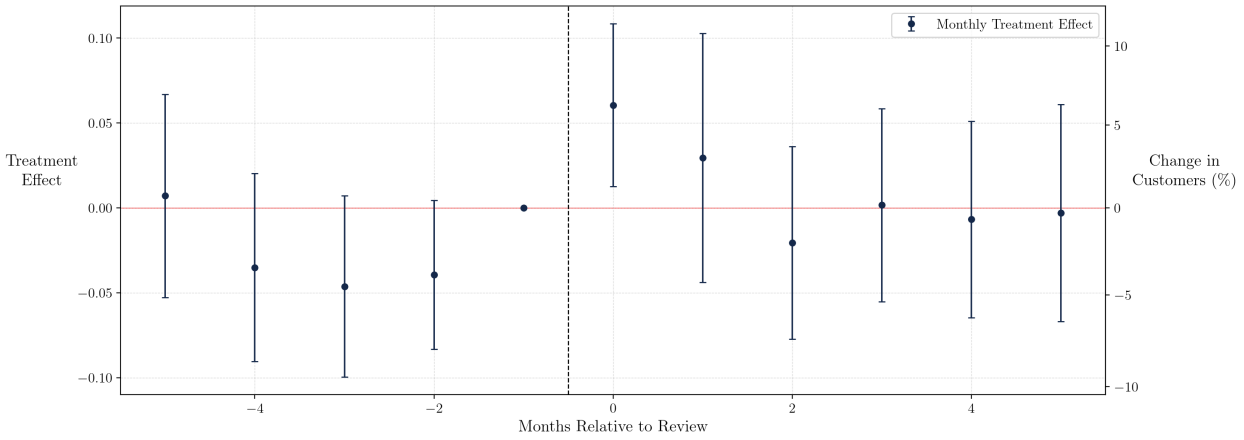
[28]Table A2 reports the coefficient estimates, where Column (1) shows the effect of any *OneBite* review. The final columns report the jointly estimated treatment effects for the heterogeneous treatment effects based on score group. Although the estimates are spread over three columns, I report them under a single column label, Column (2), as the coefficients are jointly estimated in a single regression.

treatment effects show that the average *OneBite* review creates an attention shock for the pizzeria and induces a transitory surge in customers, attention does not generate long-term growth for the firm.[29] However, the increase in customers critically depends on the review score.

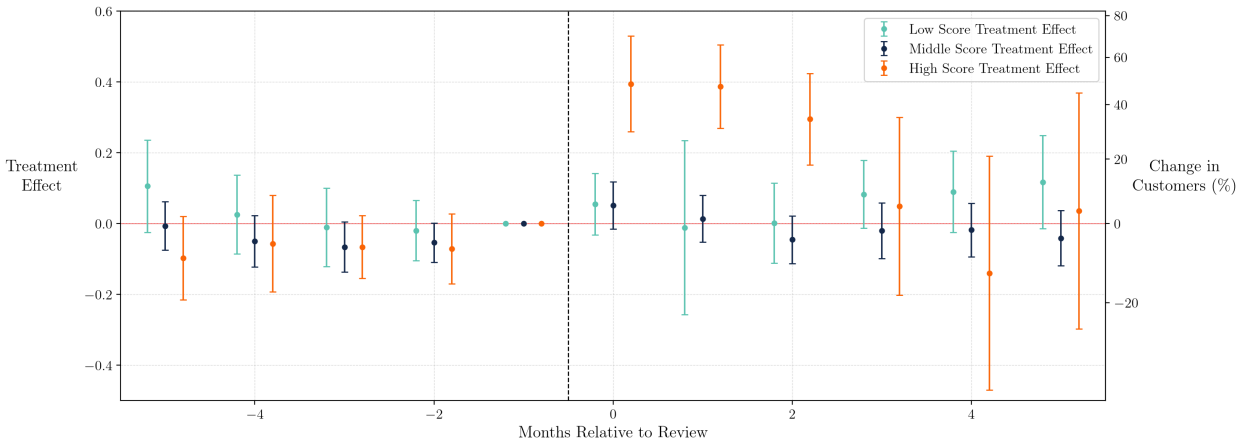
Figure 3b presents the jointly estimated treatment effects based on review score. After a *OneBite* pizza review is posted to YouTube with a score $< 7.0$ or $[7.0, 8.0)$, the internet attention does not significantly change total monthly customers at the reviewed pizzeria. However, if a pizzeria receives a score $\geq 8.0$, total customers increase by 48% in the first month following the review, and remain 47% and 34% above pretreatment levels in the second and third months, respectively. As the substantial increases in customers are significantly greater than zero and relative to all pretreatment coefficients, Figure 3b shows that the attention from a high review score draws customers to the pizzeria and improves firm performance. Aligned with the theoretical framework, high-score reviews may generate larger increases if the review leads to a positive update of a pizzeria's expected quality, generates a larger prestige effect, or draws the attention of more non-local consumers. For review scores $< 7.0$ and $[7.0, 8.0)$, the insignificant effect may result from a negative quality signal that offsets any benefit from the additional social media attention the review spurs, or the inability to generate greater awareness or prestige. Despite the significant increases in customers at pizzerias after high-score reviews, firm performance returns to its pretreatment level at pizzerias in the fourth month following the review.

To investigate why the treatment effects dissipate shortly after the review, I first evaluate whether sample attrition explains the decrease over time. However, 98.9% of treated pizzerias observed at $t = 0$ remain in the estimation sample at $t = 5$, including more than 99% of pizzerias that receive high scores. Rather, the reversion shows that attention fades and cannot generate sustained growth for a firm. When Portnoy first reviews a pizzeria, the review increases total customers as social media followers visit the viral pizzeria. The increase is highest in the initial months when consumers are most likely to see the video review on social media. However, as Portnoy posts new video reviews, older reviews become less likely to appear on social media feeds; the attention to new pizzeria reviews diverts social media followers towards the recently reviewed pizzerias. As the attention fades and new reviews divert consumers to other pizzerias or other online media trends, the attention only temporarily improves firm performance.

[29] Figure A15 presents the results for spending using a stacked event study regression and shows a similar immediate increase, but no long-term benefits.




(a) Effects of Any Review




(b) Heterogeneous Effects by Review Score

Figure 3: Total customers temporarily increase

Notes: Figure 3 presents the event study estimates from Equation 5 to show the dynamic treatment effects of receiving a *OneBite* review. Panel A shows the effects on total monthly customers in each month following treatment, whereas Panel B shows the heterogeneous treatment effects on total monthly customers by review score. The coefficient estimates are interpreted as the percentage change in each metric relative to the month prior to treatment. I correct for time and cohort heterogeneity using a stacked regression design. The confidence bars show the 95% confidence intervals with standard errors clustered at the establishment level. The left vertical axis shows the coefficient estimate from the regression, while the right vertical axis shows the percentage change interpretation using $\exp(\hat{\beta}) - 1$.

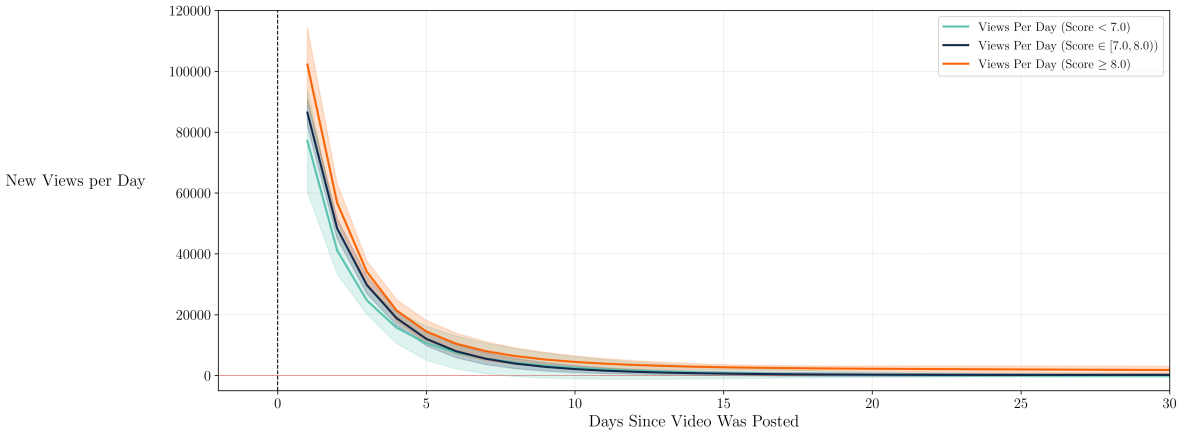



Figure 4: *OneBite* Review Views on YouTube by Score Category

Notes: Figure 4 presents the results from a generalized additive model to show how the number of new views changes over time. Data is scraped from the Internet Archive's Wayback Machine, with an average of 5.1 timestamps per video. The figure shows the temporal variation for Portnoy's three score categories. The shaded region represents a 95% confidence interval, with standard errors clustered at the review level.

Figure 4 presents evidence that the video reviews do not generate persistent online attention and plots a generalized additive model to show the non-parametric trends in new video views over time, where the vertical axis shows the estimated number of views per day and the horizontal axis shows the days since the video is uploaded. Each trend uses the listed number of YouTube views of each *OneBite* video review from Internet Archive's Wayback Machine. Figure 4 shows a rapid decline in viewership more than ten days after the review is uploaded to YouTube, at a similar rate for all score groups. Although high-score reviews are the most viewed initially, consistent with the idea that high-score reviews generate the largest attention shock, new viewership falls significantly more in less than two weeks. The rapid decrease in viewership shows that online attention is transitory, consistent with the temporary demand shock.

Despite the rapid decrease in viewership, the increase in customers for high-score reviews lasts for three months, longer than the attention on YouTube. The prolonged effect stems from several reasons. First, while consumers may not immediately visit the pizzeria after the review, when a consumer next considers dining out, the consumer's recent memory of the high-score review can influence their decision (Bordalo et al., 2020). Several months after the review, however, the consumer is less likely to recall the review and, thus, less likely to visit the pizzeria. Second, pizzerias face capacity constraints and cannot accommodate the entire surge of customers in a single month. Finally, part of the increase in later months may stem from social learning (Moretti, 2011). A consumer's decisions on where to dine may depend on information from their peers when quality is uncertain. The

aggregate demand response in later months reflects the direct effects from online attention and additional consumers who learn from their social networks, a social multiplier effect. Although the social learning explanation suggests part of the increase stems from word of mouth as opposed to the review, the increase stems directly from the attention shock and is an indirect effect on consumers. As such, a *OneBite* review may increase customers both directly from the video's viewership and indirectly through social networks.

Finally, Figure 5 presents evidence of how consumers respond to Portnoy's score classification, and that the 8.0 threshold possesses meaning to consumers.[30] The treatment effects are aggregated over the first three months following treatment for eight different treated subsamples: [0, 6.4],[6.5, 6.7], [6.8, 6.9], [7.0, 7.2], [7.3, 7.4], [7.5, 7.7], [7.8,7.9], [8.0, 10.0]. For each subsample, I restrict the sample to treated pizzerias with *OneBite* scores within the score group to determine whether the treatment effects are monotonic in score. While the left vertical axis shows the aggregated treatment effect for the score group, the right vertical axis shows the share of treated pizzerias in each group. Figure 5 shows that the average treatment effect is close to zero and statistically insignificant for subsamples with scores less than 7.8, further evidence that consumers only respond to higher-score reviews. While review scores in [7.8,7.9] generate a 17% increase in total customers in the three months following a review, pizzerias that score 8.0 or higher experience a 43% increase in total customers over the same period, which is statistically greater than the treatment effect for scores [7.8,7.9]. Although the sample lacks the statistical power to estimate a difference in discontinuities, this sharp increase indicates that consumers follow Portnoy's score classification and respond to high-score reviews.

## 6.2 Spatial Spillovers

I next investigate whether the online attention generates spatial spillovers onto nearby establishments. First, I examine whether customers substitute away from nearby alternatives. If total customers do not significantly change at nearby pizzerias after a review, then the increase in customers at the treated pizzeria does not result from business stealing from other pizzerias, and suggests the review either draws new consumers to the local area or from other establishments. Following Butts (2021), I jointly estimate the treatment effect and the spillover effects onto pizzerias within 1,000m of the review, where the control sample only includes untreated pizzerias farther than 1,000m away. Equation 7 presents the adapted specification where the coefficients of interest are $\{\beta_\ell^d\}$ and the remaining terms follow the main specification.

[30] Table A5 reports the coefficient and standard error estimates for each treatment effect.

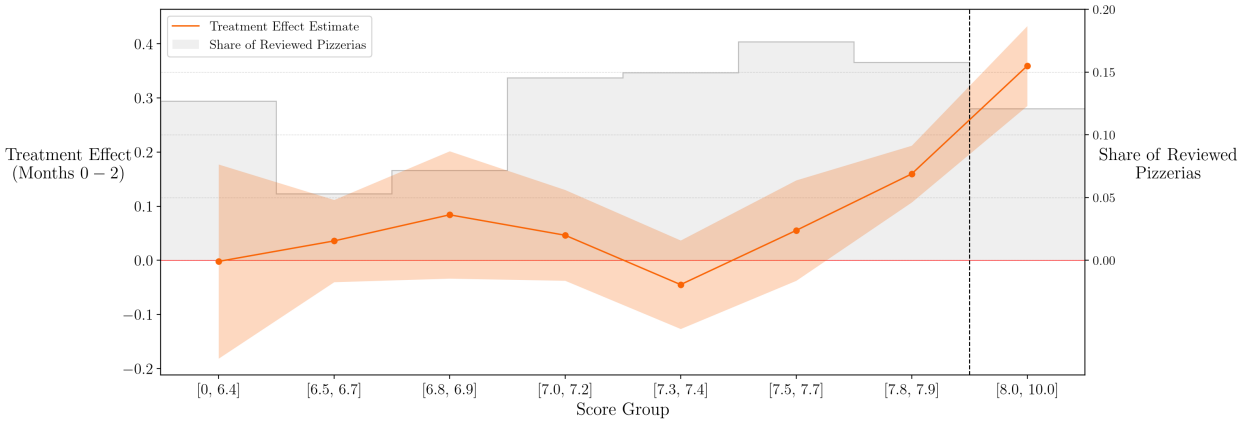



Figure 5: Treatment effects are monotonic in score

Notes: Figure 5 presents aggregated treatment effects from the first 3 months of event study estimates using a stacked regression design to correct for time and cohort heterogeneity. Each treatment effect estimate measures the change in total customers after receiving a *OneBite* review score in a specific score bin: [0, 6.4],[6.5, 6.7], [6.8, 6.9], [7.0, 7.2], [7.3, 7.4], [7.5, 7.7], [7.8,7.9], [8.0, 10.0]. The left vertical axis shows the aggregate coefficient estimate over the first 3 months, while the right vertical axis shows the share of the reviewed pizzerias within each score bin. The endpoints of the horizontal axis show the binned estimates for receiving a review score $< 6.5$ and $\geq 8.0$ respectively, while the other axis labels present the lowest score in the bin. The gold shaded region represents a 95% confidence interval with standard errors clustered at the establishment level.

$$Q_{jct} = \exp\left(\mu_j + \mu_t + \Gamma_{c(j)\times t} + \sum_{d=1}^{3}\sum_{\substack{\ell=-5 \\ \ell\neq -1}}^{5} \beta_\ell^d \mathbb{1}\left(t - \tau_j = \ell, j \in \mathcal{D}_d\right)\right) + \varepsilon_{jct}. \tag{7}$$

For each month $\ell$ after the review, Equation 7 jointly estimates three treatment effects. Figure 6 illustrates the treatment design that shows the three treatment groups. First, the treatment effect captures the change in customers at the treated pizzeria, analogous to the main specification but with a smaller control sample. The second two treatment effects are based on the treatment effects within two narrow treatment bands: pizzerias within 500m and pizzerias within a 500m to 1,000m band of the review. If customers substitute towards the reviewed pizzeria, the spillover effects onto nearby pizzerias will be statistically negative. Since Equation 7 estimates the increase in customers at the treated pizzeria compared to pizzerias outside of all treatment bands, the estimated treatment effect does not violate SUTVA if consumers do not substitute away from pizzerias more than 1,000m away. Since customers only respond to high-score reviews, I focus on spillovers of high-score reviews and report the results for any review in Appendix A.

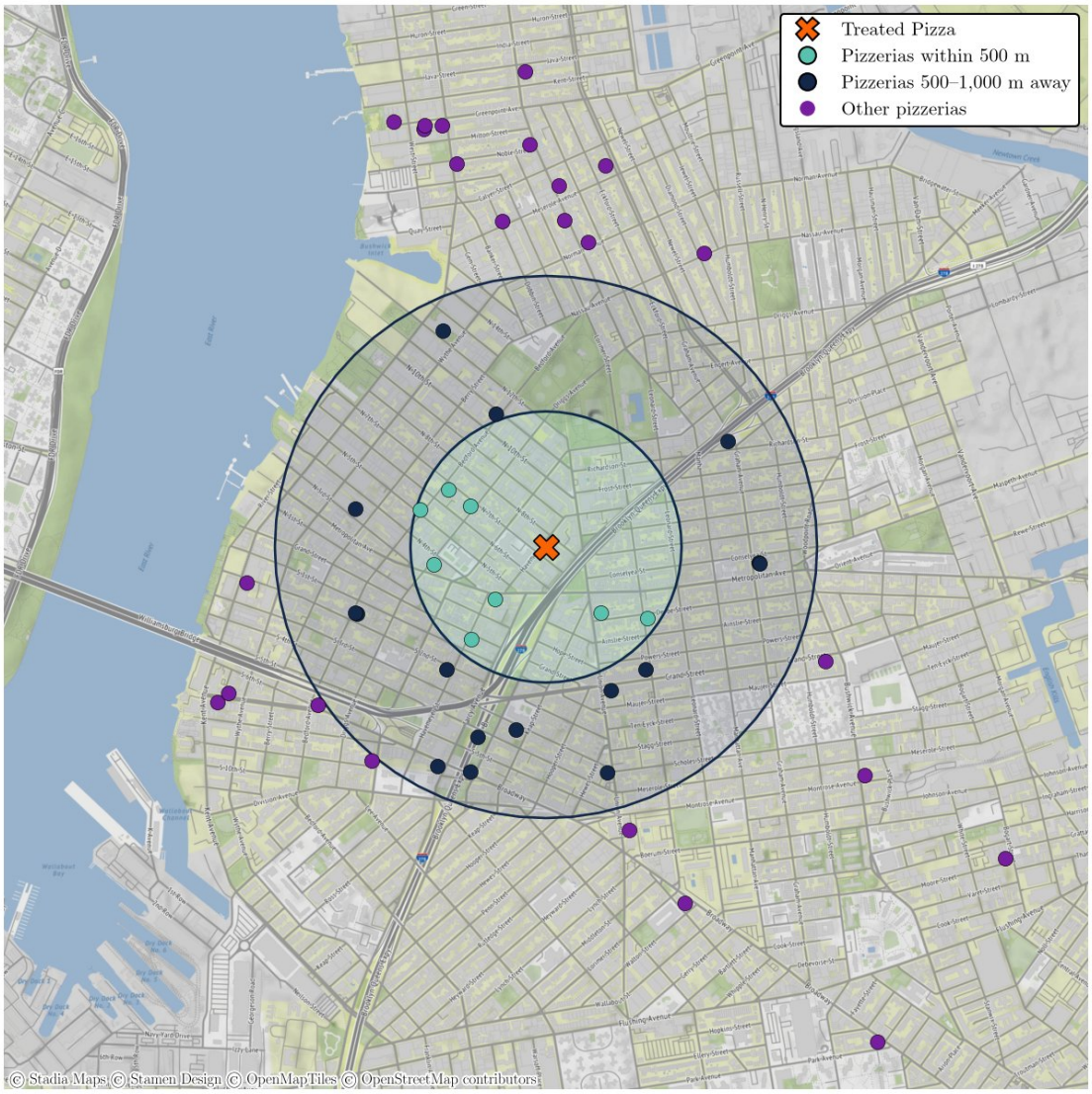



Figure 6: Spatial Spillovers onto Nearby Pizzerias

Notes: Figure 6 illustrates the spatial difference-in-differences research design following Butts (2021). The red circles represent other establishments. Establishments are assigned to two treatment rings around the reviewed pizzeria, within 500m and between 500m and 1,000m. The treatment effect is on the establishment, marked with an 'x', and is jointly estimated with the treatment effect estimates on all establishments within the two treatment rings. Establishments in the same NAICS code outside of 1,000m serve as the control group.

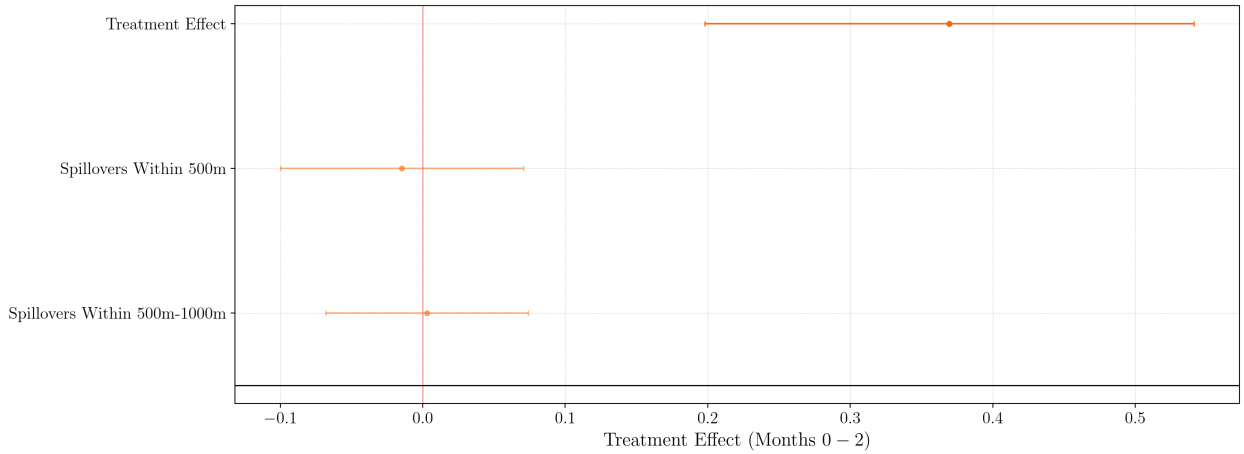



Figure 7: Limited Evidence of Substitution after a High-Score Review

Notes: Figure 7 presents the effects on total customers over the first three months following the posting of a high-score *OneBite* review (score $\geq 8.0$) to YouTube. The direct treatment effect on the reviewed pizzeria and spatial spillovers onto other pizzerias within 500 m and between 500 m and 1,000 m are jointly estimated following Equation 7. I correct for time and cohort heterogeneity using a stacked regression design. The confidence bars show 95% confidence intervals, with standard errors clustered at the establishment level. Table A7 reports the coefficient estimates for both the effects of a high-score review and of any review.

I estimate how a *OneBite* review affects nearby pizzerias using a stacked PPML event study design, analogous to the main specification. Figure 7 presents the treatment effect over the first three months after a review.[31] After the attention shock to the pizzeria, total customers at competing pizzerias within 500m decrease by 1%, but increase by 0.3% at pizzerias between 500m and 1,000m away. While the negative point estimate for pizzerias within 500m suggests that some substitution may occur, the specification does not identify a significant effect. Furthermore, the treatment effect on the treated pizzeria increases to 45% over the first three months following a review, marginally greater, though statistically indistinguishable from the main specification. If consumers substitute away from nearby pizzerias, excluding nearby pizzerias from the control group would cause the treatment effect to attenuate. Instead, the point estimate increases marginally. Although the spillover effects are not a precise null, the near-zero and statistically insignificant spillover effects, combined with the marginal increase in treatment effect, do not provide any evidence that the treatment effect results from business stealing from nearby pizzerias.

Although the attention does not lead to substitution away from other pizzerias, the attention may generate spatial spillovers to local establishments in other industries. If nearby establishments provide complementary goods and services, the increase in local

[31] Figure A8 presents the treatment and spillover effects of any review. Figure A9 presents the corresponding effects of any review and high-score reviews, restricting the analysis to the first month after the review.

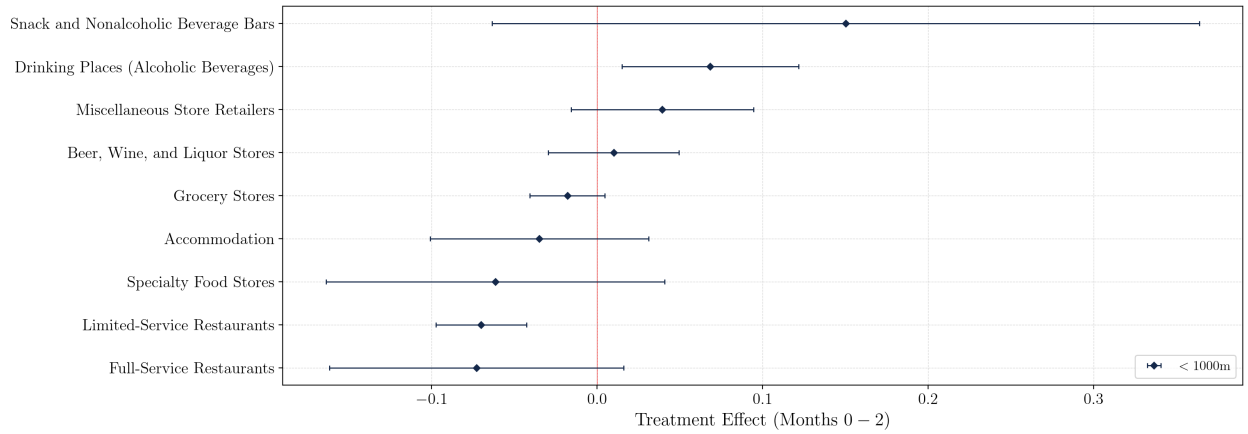



Figure 8: Spillovers across Industries

Notes: Figure 8 presents the average treatment effect over the first three months following a high-score *OneBite* review (months 0 to 2 in the event study) on total monthly customers at nearby establishments, estimated separately by the industry of the nearby establishment, where industries are defined by NAICS code. Treatment is restricted to reviews with a score $\geq 8.0$. The sample includes all establishments within 1,000 m of any reviewed pizzeria in SafeGraph's Spend Patterns data, where reviewed pizzerias need not be matched. I correct for time and cohort heterogeneity using a stacked regression design and normalize $t = -1$ to 0. The confidence bars show 95% confidence intervals adjusted for multiple hypothesis testing using $q$-values, with standard errors clustered at the establishment level. Table A4 reports the coefficient estimates.

foot traffic may increase firm performance, consistent with consumption-based agglomeration. However, establishments that provide substitutes may lose business if consumers redirect attention towards the reviewed pizzeria. I estimate how customers at nearby retail establishments within 1,000m of the treated pizzeria change after a *OneBite* review, analogous to the main specification. Unlike the previous spillover effects, the specification does not directly estimate the treatment effect at the treated pizzeria and only includes establishments in the same 4-digit NAICS code. This allows my treatment sample to include the full sample of *OneBite* reviews on the app and YouTube and not only pizzerias matched to SafeGraph. Figure 8 presents the spillover effect estimates from a high-score review, where the confidence intervals show FDR-sharpened $q$-values to control the false discovery rate that arises from multiple hypothesis testing.

For high-score reviews, the spillover effects present a similar narrative, as customers increase at "Drinking Places (Alcoholic Beverages)" (+7%) within 1,000m and decrease at "Limited-Service Restaurants" (−7%).[32] The pattern reveals the distinct complementary and substitution patterns for consumers. The decrease at limited-service restaurants indi-

[32]Figure A10 presents the average treatment effect over the first three months after any review. Although the results attenuate, the narrative remains similar, with a marginal increase in "Beer, Wine, and Liquor Stores" (+2%) within 1,000m of a reviewed pizzeria and a marginal decrease in "Limited-Service Restaurants" (−1%).

cates that customers in the area substitute away from alternative fast-food and takeout dining options.[33] While direct spillovers onto other pizzerias show scant evidence that consumers substitute away from nearby pizzerias, the decrease in customers at limited-service restaurants suggests that customers will substitute away from other fast-food cuisines. However, the increase in establishments that serve alcoholic beverages indicates that consumers drawn to the area view the drinking establishments as complements to the reviewed pizzeria, providing a positive demand shock.

The demand spillovers from a *OneBite* review provide direct evidence of a local demand externality, a mechanism that defines consumption-based agglomeration. Rather than production-focused explanations where firms co-locate for greater returns in production or lower marginal costs, consumption-based agglomeration mechanisms show that firms that share a customer base should cluster to benefit from demand spillovers. The increase in customers at "Drinking Places (Alcoholic Beverages)" that follows a high-score review in the area provides evidence of this mechanism. The attention to a single pizzeria attracts consumers to the surrounding retail cluster and generates demand externalities for establishments in the area that provide desirable products to the new consumers. I further investigate this claim and examine changes in customer composition at reviewed restaurants. Using individual-level Google review data from Li et al. (2022), I collect all available reviews posted by customers who reviewed a high-score pizzeria in the three months before the *OneBite* review and compare the users to those who posted reviews of the treated pizzeria in the three months afterward. The Google reviews identify the type of establishments each group reviews, and therefore, visits. Table A6 reports the share and number of reviewers in each group who reviewed different establishment types.[34] The difference in shares indicates that the users who leave Google reviews in the three months after the review are less likely to review bars, counterintuitive to the positive increase in customers at "Drinking Places (Alcoholic Beverages)." However, the number of pizzeria reviewers who also review bars increases by 9.8%, consistent with the positive spatial spillovers for complementary establishments to provide evidence of consumption-based agglomeration.

[33]The spillovers onto "Limited-Service Restaurants" and "Full Service Restaurants" exclude all control pizzerias from the *OneBite* app and any establishment that contains "pizza" or "pizzeria" in its name to mitigate any effects from substitution away from other pizzerias. Although the sample may still contain pizzerias, this exclusion limits the group to mostly non-pizzerias.

[34]Although Google distinguishes over 4,000 establishment categories, I aggregate these into 14 broad categories, combining detailed classifications such as different types of clothing stores. I retain area pizza restaurants as a separate category to compare reviewers' engagement with other pizzerias.

## 6.3 Local versus Non-Local Consumers

While the significant increase in customers suggests that a *OneBite* review affects consumer decisions, I next investigate if effects on specific consumers may vary based on a consumer's prior knowledge of the local market. Consumers may learn about pizzeria quality either through search or through direct experience (Nelson, 1970). As a visit to a pizzeria is a relatively low-cost good, local consumers may view pizzerias as an experience good because of the ample opportunities to purchase a variety of products in the market and learn about the pizzeria's quality through consumption. However, non-local consumers either do not frequent the area or face commute costs to learn through experience. The fewer opportunities mean that non-local consumers rely on search, including online reviews. Furthermore, advertising's primary effect is informing consumers (Ackerberg, 2003), and product information primarily affects consumers with no prior experience with the brand (Ackerberg, 2001). If *OneBite* reviews generate attention and convey information, the effects may be larger for non-local consumers who possess less prior knowledge of the local market and face an awareness shock about the product.

To investigate the role of experience in the market, I estimate how the treatment effect varies with the distance between a consumer's home location and the reviewed pizzeria. I use distance to proxy for market familiarity and assume that consumers who live closer to the pizzeria possess greater knowledge of the local market. I use SafeGraph's records of the total count of customers from each city, geocode each city to its county, record its distance to the pizzeria's county centre, and group customers into five distance bins: within 5km, between 5km and 10km, between 10km and 25km, between 25km and 50km, and beyond 50km. Following the main specification, I estimate how a *OneBite* review affects the total number of customers from each distance bin using a stacked PPML event study design. For the distance specifications, PPML provides an additional benefit to account for any zero counts in some distance bins.[35] Figure 9 presents the average effects in the first three months following a high-score review.

Figure 9 shows that the effects of a high-score review increase with distance from the pizzeria. After a pizzeria receives a high-score review, consumers from counties within 5km of the treated pizzeria's county, including the same county, increase by 19% over the first three months. The consumer response increases with distance, with a 27% increase in consumers between 5km and 10km, a 32% increase for consumers between 10km and

[35]Some consumers in SafeGraph's underlying sample may not report their home city, so the sum of the customers across distance bins is less than the total customer count in the main specification.

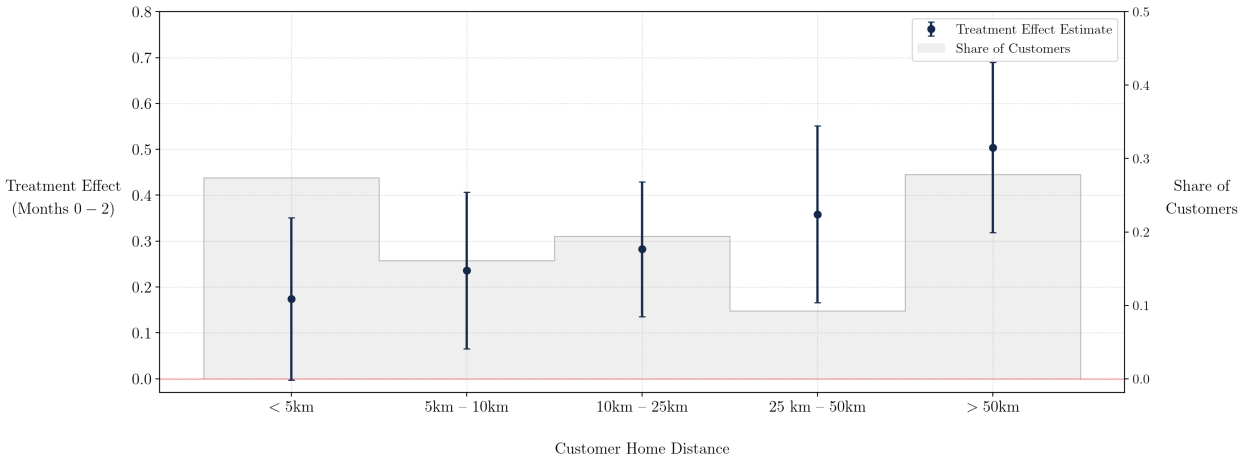



Figure 9: Treatment Effects by Consumer Distance

Notes: Figure 9 presents the aggregated treatment effects of a high-score *OneBite* review on total monthly customers in the first three months following the review, estimated separately by the distance between the customer's home location and the reviewed pizzeria. The five categories group customers living within 5km, between 5km and 10km, between 10km and 25km, between 25km and 50km, and beyond 50km. I correct for time and cohort heterogeneity using a stacked PPML event study design and normalize $t = -1$ to 0. The confidence bars show the 95% confidence intervals with standard errors clustered at the establishment level. Table A3 reports the coefficient estimates for each treatment effect.

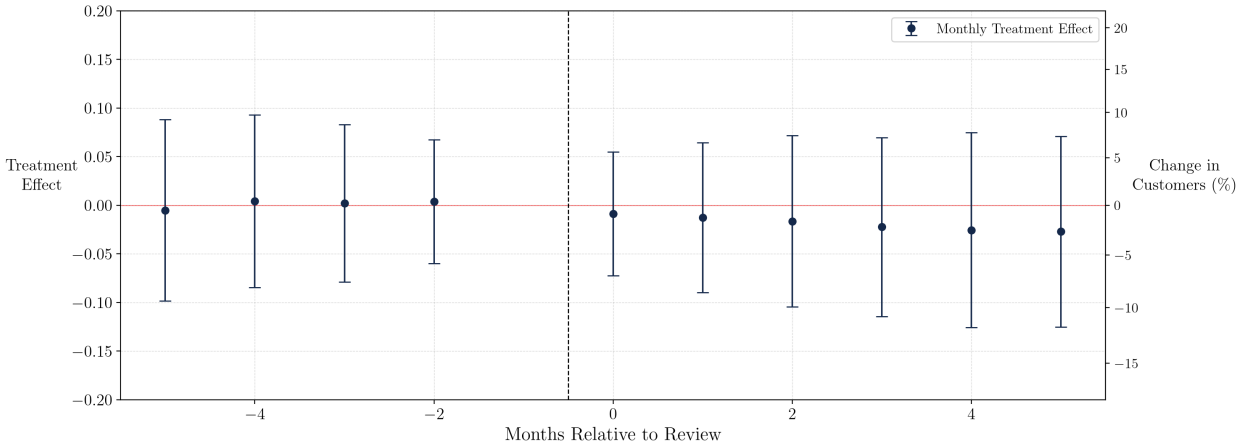
25km away, and a 43% increase in consumers between 25km and 50km away. The largest proportional increase is from consumers who live in counties more than 50km away, who increase by 66%. The monotonic relationship between distance and the consumer response is consistent with the premise that consumers with less prior knowledge of the market will rely on product information acquired through search rather than experience and will respond more to online attention. Furthermore, the increase in non-local customers is not a mechanical result of a lower sample mean. The right vertical axis of Figure 9 shows the average share of customers at each distance bin in the untreated sample. Although the bins vary in size across the distance measures, the total number of customers from counties within 5km of the treated pizzeria (27.3%) is nearly identical to the number of customers from counties more than 50km away (27.9%). Rather, the greater increase among non-local customers is consistent with the review increasing awareness and generating a larger response for consumers with less prior information about the market. The differential response from non-local consumers directly motivates the product awareness channel in my theoretical framework. If non-local consumers are unequally aware of all the products in the market, an attention shock will increase their awareness of the product and generate a larger response relative to local consumers. Although the increase in customers at narrow distance bins indicates that local consumers respond to the online attention, the effect increases with distance, which suggests that product awareness contributes to the consumer

response.

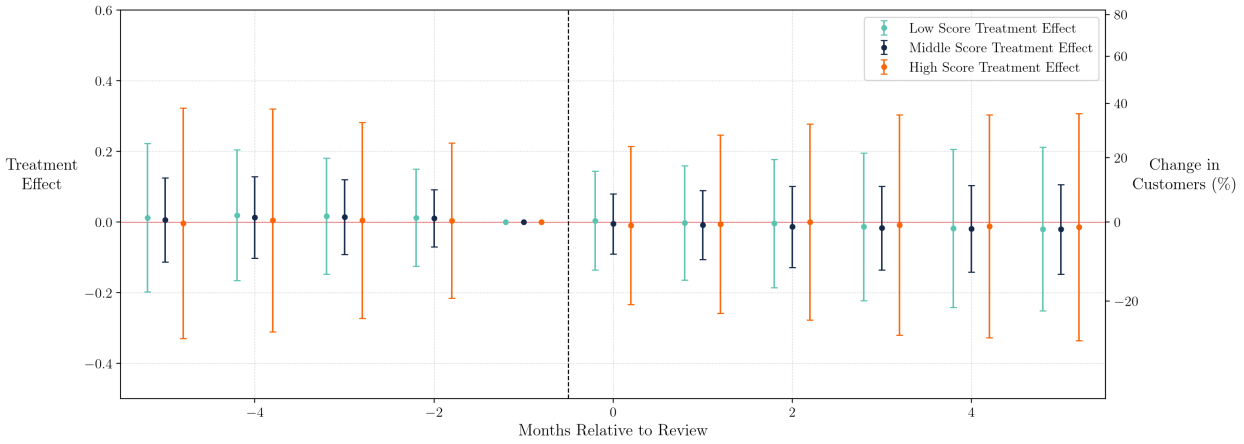
## 6.4 Robustness Checks

I perform several robustness checks to validate the estimated treatment effect. First, I run a permutation placebo that randomly reassigns the observed treatment dates and review scores across pizzerias to test whether an idiosyncratic allocation of review dates and scores across pizzerias explains the estimated treatment effects. Figure 10a presents the average treatment effect from any review, while Figure 10b presents the heterogeneous treatment effects by score category. Across 1,000 permutation iterations, both permutation placebos show that the estimated effects are centred on zero and statistically insignificant, which indicates that the event study specification does not mechanically generate treatment effects at randomly assigned review dates. Furthermore, the 6.2% increase in customers in the first month following the review exceeds the placebo estimates in over 99% of the permutation iterations, and the 43% increase from a high-score review over the first three months exceeds the placebo estimates in every iteration. Since the treatment effect exceeds almost all placebo estimates in both specifications, the estimated treatment effects are unlikely to arise by chance and are attributable to the *OneBite* review.

Next, I conduct a sensitivity analysis following Rambachan and Roth (2023) to assess how sensitive the estimates are to post-treatment violations of parallel trends. With high-score reviews, many of the pre-treatment coefficients are negative and gradually increase over time. Although no coefficient is statistically different from zero, the increase over time indicates an, albeit subtle, increase in customers before the *OneBite* review. To test whether the trend influences the treatment effect estimates, Figure 11 shows the results of the sensitivity analysis for the effects of any review and the effects of a high-score review if the violation of parallel trends in the post-treatment period is $\bar{M}$ times larger than the maximum pre-treatment violation for different values. Figure 11 presents four panels showing the sensitivity of the estimated effects of any review and a high-score review, both in the first month following the review and aggregated over the first three months. Although Figures 11a and 11b show that the average effect of the review becomes statistically insignificant when the post-treatment violation equals the largest pre-treatment violation, Figures 11c and 11d show that the effects of a high-score review remain statistically significant under the same restriction. The sensitivity analysis, therefore, shows that only high-score reviews are robust to violations of parallel trends, which reinforces that high-score reviews drive the overall increase in customers.



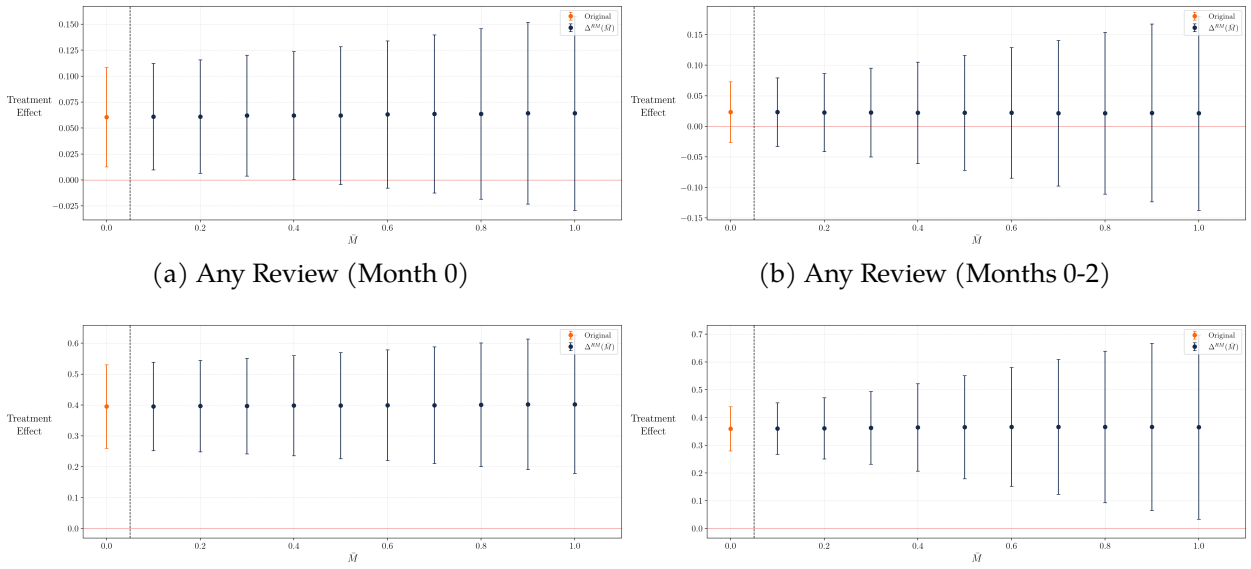

(a) Randomly Assigned Treatment Date




(b) Randomly Assigned Treatment Date and Score

## Figure 10: Permutation Placebo Tests

Notes: Figure 10 presents two placebo tests to validate the research design, where both figures report event study estimates for total monthly customers and correct for time and cohort heterogeneity using a stacked regression design and normalize $t = -1$ to 0. Figure 10a presents the event study estimates from a permutation test across 1,000 iterations, where each iteration randomly assigns the actual distribution of treatment dates to a random subset of untreated pizzerias and re-estimates Equation 5 on the untreated sample. Figure 10b presents an analogous specification that assigns the score of the review and re-estimates the heterogeneous treatment effects. The points show the mean placebo estimate at each event time and the bars show the $2.5^{\text{th}}$ to $97.5^{\text{th}}$ percentiles of the permutation distribution.




(a) Any Review (Month 0)

(b) Any Review (Months 0-2)

(c) High-Score Review (Month 0)

(d) High-Score Review (Months 0-2)

Figure 11: Honest Difference-in-Differences

Notes: Figure 11 presents the results from an honest difference-in-differences specification following Rambachan and Roth (2023). $\bar{M}$ denotes how much larger a post-treatment violation of parallel trends may be relative to the largest violation observed before treatment. Figure 11a presents the treatment effect for any *OneBite* review in the first month after the review is posted to YouTube, while Figure 11b presents the results for the aggregated treatment effect over the first 3 months following the review. Figures 11c and 11d present the analogous results for *OneBite* reviews with scores $\geq 8.0$.

To assess whether the main specification violates SUTVA, I omit control pizzerias located within a 500m or a 1,000m radius of a treated pizzeria and re-estimate Equation 5. One concern is that if customers substitute away from untreated pizzerias, total monthly customers at untreated restaurants will decrease and the main specification will overestimate the true treatment effect from a review. The exclusion creates an identical control group to the spatial spillovers specification, only without jointly estimating spillover effects. Figure A11a presents event study results for any *OneBite* review where the control sample omits pizzerias within 500m, while Figure A11b presents the results after the sample omits pizzerias within 1,000m. After I omit control pizzerias within a 500m radius of the treated pizzeria, the increase in total customers during the first month increases to 6.5%. While excluding control pizzerias within a 1,000m radius decreases the estimates to 6.4%, both results are above and statistically similar to the main results. Furthermore, Figures A11c and A11d show the analogous results for the heterogeneous treatment effects and show that customers continue to increase after high-score reviews, with an average increase of 44% in the first three months when the specification omits pizzerias within 500m and 45% when the specification omits pizzerias within 1,000m. Since all treatment effect estimates increase after the specification omits nearby pizzerias, the event studies in Figure A11 show no evidence of bias from substitution, evidence that the main specification does not violate the SUTVA.

Finally, I re-estimate the treatment effects using different corrections for treatment effect heterogeneity across cohorts and report the results in Figure A12. Since the corrections use linear specifications, I use the logarithm of total monthly customers as the dependent variable. The log-linear specification changes the parallel trends assumption, which now assumes that average log customers follow similar trends, rather than following similar proportional changes as in the PPML specification. For the average *OneBite* review, the linear specifications show a moderate increase in customers in the first two months following the review that ranges between +5% and +11%. While not all corrections provide statistically significant estimates in each month, the estimates for each specification are jointly positive over the first two months. Similar to the PPML model, low-score and middle-score reviews do not generate any increase in customers. High-score reviews continue to explain the overall treatment effect, with increases that range between 32% and 50% over the first two months. Although the temporal decay is relatively quicker in the log-linear specifications, the central narrative remains: high-score *OneBite* reviews generate a significant increase, but the attention shock dissipates quickly.

# 7 Decomposition

## 7.1 Estimation

Following my theoretical framework, I estimate a triple-difference specification and map the reduced-form coefficients to their structural parameters. Equation 8 presents the empirical counterpart to Equation 3, where $Q_{jgct}$ measures the total customers of type $g$ who visit pizzeria $j$ in county $c$ and month $t$. I define non-local consumers as consumers whose home county's centre lies more than 50km from the centre of the pizzeria's county. Equation 8 includes consumer type-pizzeria fixed effects $\mu_{jg}$, consumer type-county-time fixed effects, and treatment terms that capture information ($\beta^I$), prestige ($\beta^P_{\sigma\ell}$), and awareness ($\beta^A_{\sigma\ell}$). I define the information update as the difference between the *OneBite* score $s_j$ and the rescaled Google rating $\widetilde{R}_j$. To directly compare Google's 5-star rating with Portnoy's 10.0-point scale, I rescale Google ratings to match the mean and standard deviation of Portnoy's scores.[36] After the transformation, 83% of high-score reviews provide a positive information update to consumers, compared to 56% of middle-score reviews and 27% of low-score reviews.

$$\ln Q_{jgct} = \mu_{jg} + \Gamma_{g\times c(j)\times t} + \overbrace{\beta^I(s_j - \widetilde{R}_j)}^{\text{Information}}\mathbb{1}(t \geq \tau_j) + \sum_{\sigma\in\{H,M,L\}}\sum_{\substack{\ell=-5\\ \ell\neq -1}}^{5}\overbrace{\beta^P_{\sigma\ell}}^{\text{Prestige}}\mathbb{1}(t=\tau_j+\ell, s_j=\sigma)$$
$$+ \sum_{\sigma\in\{H,M,L\}}\sum_{\substack{\ell=-5\\ \ell\neq -1}}^{5}\underbrace{\beta^A_{\sigma\ell}}_{\text{Awareness}}\mathbb{1}(g=N)\mathbb{1}(t=\tau_j+\ell, s_j=\sigma) \tag{8}$$

As in my main specification, I estimate Equation 8 using a stacked PPML design to recover the reduced-form parameters $\{\beta^I, \beta^A_{\sigma\ell}, \beta^P_{\sigma\ell}\}$ for each $\ell$ and $\sigma$. I then jointly estimate the structural parameters $\Theta = (\eta, \phi_L, \phi_M, \phi_H, \kappa_L, \kappa_M, \kappa_H, \delta_p, \delta_a)$ using a minimum distance estimator on the post-treatment reduced-form estimates. Table 2 reports the structural estimates and reduced-form relationships that identify each parameter. The overidentification test yields a $J$ statistic of 34 with 28 degrees of freedom ($p = 0.190$) and fails to reject the restrictions of the theoretical framework.

For each parameter estimate, $\theta$, I calculate the percentage change in expected cus-

[36] Figure A13 presents a heatmap to illustrate how the information update varies across review scores.

Table 2: Structural Parameter Estimates

| Parameter | Identified by | Estimate | Std. Error | 95% Confidence Interval |
|---|---|---|---|---|
| *Panel A: Information* | | | | |
| $\eta$ | $\beta^I$ | $-0.005$ | $0.011$ | $[-0.026,\ 0.017]$ |
| *Panel B: Prestige* | | | | |
| $\phi_L$ | $\beta^P_{L0}$ | $0.059^{*}$ | $0.034$ | $[-0.008,\ 0.127]$ |
| $\phi_M$ | $\beta^P_{M0}$ | $0.033$ | $0.029$ | $[-0.023,\ 0.089]$ |
| $\phi_H$ | $\beta^P_{H0}$ | $0.368^{***}$ | $0.041$ | $[0.288,\ 0.448]$ |
| $\delta_p$ | $\dfrac{\beta^P_{\sigma,\ell+1}}{\beta^P_{\sigma\ell}}$ | $0.690^{***}$ | $0.121$ | $[0.454,\ 0.927]$ |
| *Panel C: Awareness* | | | | |
| $\kappa_L$ | $\beta^A_{L0}$ | $0.009$ | $0.051$ | $[-0.092,\ 0.110]$ |
| $\kappa_M$ | $\beta^A_{M0}$ | $-0.011$ | $0.038$ | $[-0.085,\ 0.063]$ |
| $\kappa_H$ | $\beta^A_{H0}$ | $0.331^{***}$ | $0.079$ | $[0.177,\ 0.485]$ |
| $\delta_a$ | $\dfrac{\beta^A_{\sigma,\ell+1}}{\beta^A_{\sigma\ell}}$ | $0.651^{***}$ | $0.191$ | $[0.276,\ 1.025]$ |
| Overidentification $J$ statistic | | | | $34.327$ |
| Degrees of freedom | | | | $28$ |
| $p$-value | | | | $0.190$ |

Notes: Table 2 presents structural parameter estimates and their reduced-form mappings. Structural parameters are estimated by minimum distance using reduced-form coefficients from a stacked PPML specification with consumer type-pizzeria and consumer type-county-time fixed effects. Score groups are indexed by $\sigma \in \{H, M, L\}$, corresponding to scores $\geq 8.0$, $[7.0, 8.0)$, and $< 7.0$, respectively. The overidentification $J$ statistic tests the restrictions imposed by the structural model. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

tomers as $100[\exp(\theta) - 1]$, holding the other channels constant. Panel A in Table 2 shows that the information channel does not significantly explain the consumer response. Although the point estimate is negative, the parameter's low absolute value and narrow confidence band suggest consumers do not respond to the information and quality update from a review. Holding prestige and awareness at their estimated values for a high-score review, an information update at the $90^{\text{th}}$ percentile of the distribution shifts consumer demand by only 1.6 percentage points. The response is smaller for both low-score and middle-score reviews.

While the information update does not explain the consumer response, Panel B shows that prestige generates a significant consumer response. The prestige channel captures the value-added of an internet personality that is unrelated to product quality and brand awareness. Although Panel B shows middle-score reviews do not generate prestige, the prestige from low-score reviews increases expected customers by 6.1%, holding all else equal. Previous research shows that negative publicity can increase sales by increasing product awareness (Berger et al., 2010). However, this increase suggests that negative publicity creates value, which may reflect how the entertainment value of negative video reviews passes through to the value of visiting the pizzeria. While low-score reviews generate a marginal increase, prestige from a high-score review increases expected customers by 44.5%, reflecting the consumption value that Portnoy's score classification creates. The estimates show how digital media redirects consumer attention and influences consumer choices by creating value from visiting a trending establishment, independently of product awareness and expected quality. Although the initial prestige effect is substantial, prestige retains only 69% of its value each month following the review as viral attention fades.

Finally, Panel C shows that high-score reviews increase expected customers among non-local consumers by 39.2%, while low-score and middle-score reviews generate no significant awareness response. Although the effect is smaller relative to prestige, the results indicate that attention from an independent source can likewise increase product awareness and influence consumer demand. This consumer response echoes the advertising literature that shows advertising works through awareness more than information (Honka et al., 2017; Clark et al., 2009). However, unlike the prestige response to low-score reviews, negative publicity may not generate a demand response from increased product awareness.

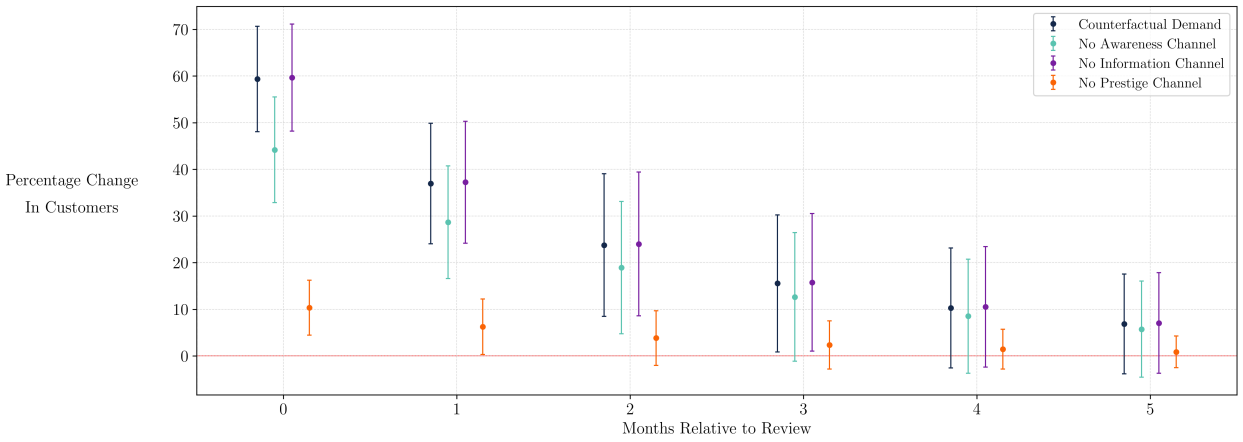



Figure 12: Consumer Response Counterfactuals

Notes: Figure 12 presents the estimated consumer response following a high-score review. The four trends show the predicted consumer response with all channels active, the response when the review does not communicate any information, the response when the review does not change product awareness, and when the review does not generate prestige. The counterfactual consumer responses are constructed using the structural parameter estimates and baseline customer counts for each consumer type, averaged over the five months before the review. Effects are expressed as percentage changes in total customers relative to baseline. The error bars represent 95% confidence intervals with standard errors calculated using the delta method.

## 7.2 Counterfactuals

I conduct three counterfactuals using the structural parameter estimates to examine how each channel contributes to the consumer response to a high-score review. Each counterfactual removes one channel while holding the others fixed. To construct the baseline response, I calculate the average number of customers for each consumer type over the five months before a *OneBite* review and use the structural estimates to predict demand each month afterward. The three counterfactuals I investigate independently omit one of the channels. Figure 12 compares the predicted consumer response with all three channels active against the three counterfactual demand paths.

Figure 12 shows that the model predicts a 59% increase in customers in the first month following the review, larger than the 48% increase in the reduced-form. However, the decay is faster, as the model predicts increases of 37% and 24% in the subsequent two months, lower than the 47% and 34% increases in the main results. The disparity reflects the model's geometric decay restrictions on prestige and awareness, while the reduced-form specification allows the response to vary freely across months. However, both approaches show the treatment effect's decay with no sustained growth.

For the first counterfactual, I examine the consumer response when the review does not change product awareness. Under this restriction, the consumer response comes only

from the information update in the review and prestige, both of which enter a consumer's utility. Figure 12 shows that without a change in product awareness, the increase in consumer demand falls to +44% in the first month after the review, a 25% reduction in the initial consumer response. The decrease shows that a *OneBite* review is similar to an unpaid advertisement that increases consumer awareness about the product to attract additional customers. Although previous research emphasizes advertising's role in increasing brand awareness, the substantial response that remains without this channel shows that consumers also respond to other mechanisms.

Second, I investigate how the consumer response changes when a review does not communicate product quality information to consumers. When I restrict the information channel, the consumer response remains nearly identical to the response when all three channels are active. One possible confound is that the prestige channel may partly capture the information channel signal if consumers interpret Portnoy's high score as a direct quality signal. However, if the review reveals a pizzeria's quality to consumers, high-score reviews should generate a persistent effect as consumers learn about a pizzeria's true quality. Following the structural definitions, any persistent effect is captured by the information update, which is positive for most high-score reviews. This suggests that if consumers respond to the specific quality update in the review, the update should generate a persistent demand response that appears in the information channel. However, both the reduced-form estimates and the model predictions show a temporary increase in customers. The nearly unchanged counterfactual path and the absence of a persistent response show that product quality information does not significantly contribute to consumer response.

Finally, I examine the consumer response without prestige. This counterfactual, in essence, treats the review as an advertisement, which only signals quality information and increases product awareness. Under this counterfactual, the model predicts a 10% increase in customers in the first month after the review, an 83% reduction in the initial consumer response. The results from the counterfactual demand show the value added of internet personalities in the media. Although firms investing in advertising can increase customers through increased product awareness, the media generates significant consumption value that incentivizes consumers to purchase specific products and participate in online trends, well beyond the value of product information or awareness.

# 8 Conclusion

In the era of information abundance, scarce consumer attention governs consumer choice. In this paper, I investigate how attention from a *OneBite* review on social media affects consumer decisions. Exploiting the quasi-random variation in the review's upload date, I find that high-score reviews increase customers at the establishment for three months. However, low-score and middle-score reviews do not significantly increase customers for the establishment. The influx of customers following a high-score review induces a local demand shock that benefits nearby drinking places. Despite the significant increase, viral attention fades and firm performance returns to its pretreatment levels within four months of the review. Although a shift in brand awareness explains part of the consumer response, I find no significant response to quality information. Instead, the prestige from visiting a trending pizzeria explains the majority of the consumer response and highlights the importance of internet personalities in the modern digital economy. Future research should aim to separate the direct effect of attention and the indirect effects from a social multiplier. Although previous literature documents the social multiplier effect through information (Moretti, 2011), understanding how social interactions amplify social media prestige is essential to assess the value-added of the media, and whether attention on social media can compound to create sustained growth for an establishment.

Although this paper focuses on a single attention shock, the results provide broader insights for both establishments and policymakers. For establishments, the findings show that consumers immediately respond to social media attention, which can generate a temporary surge in firm performance. With the rise of internet influencers and the creator economy, the estimates show the consumption value that internet personalities add to products through their endorsements, independent of any product information and awareness. However, the temporary response shows that attention on digital media is not sufficient for sustained growth. For policymakers, the transitory response shows how attention on social media can rapidly alter behaviour. Previous research shows that media influences behaviour beyond consumption decisions. With the continued prevalence of social media in everyday life, the results highlight the importance of understanding how individuals respond to online personalities, as social media can induce an immediate behavioural change.

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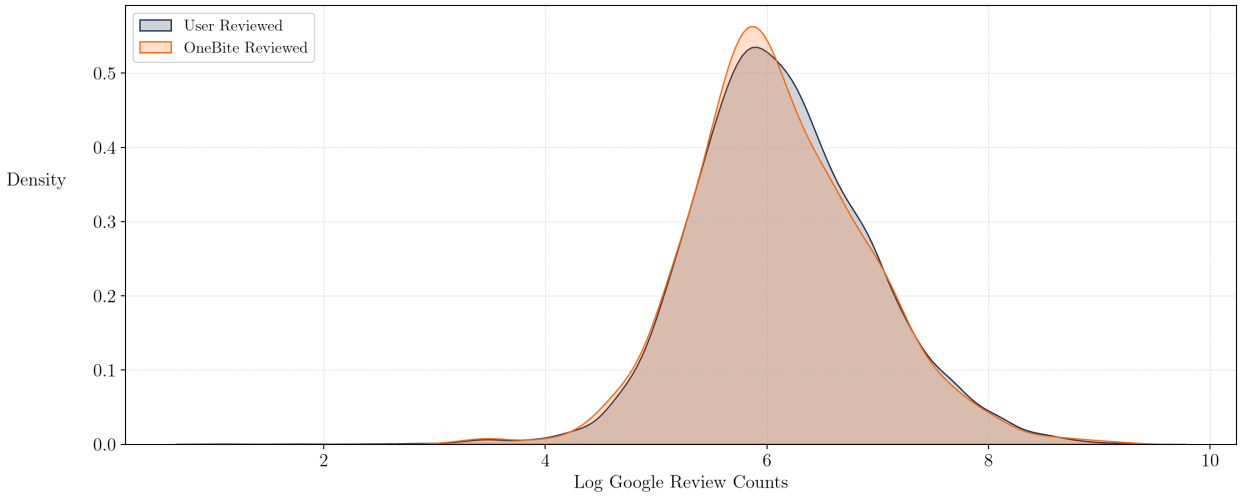
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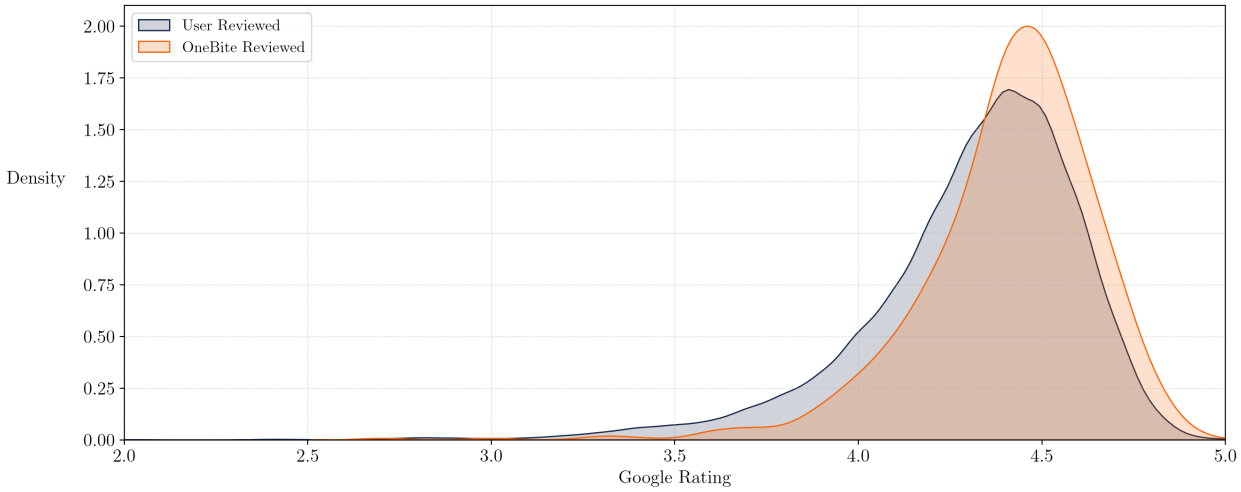
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# A Additional Tables and Figures




(a) Log Review Count




(b) Average Google Rating

Figure A1: Google Ratings for Treatment and Control Pizzerias

Notes: Figure A1 presents two kernel density plots to compare the Google ratings of pizzerias that receive a *OneBite* review to pizzerias that app users reviewed but did not receive a review. Figure A1a shows the kernel density of Log Review count to compare how frequently consumers review each restaurant type, while Figure A1b plots the average rating across the two groups, taken at a single point in time.

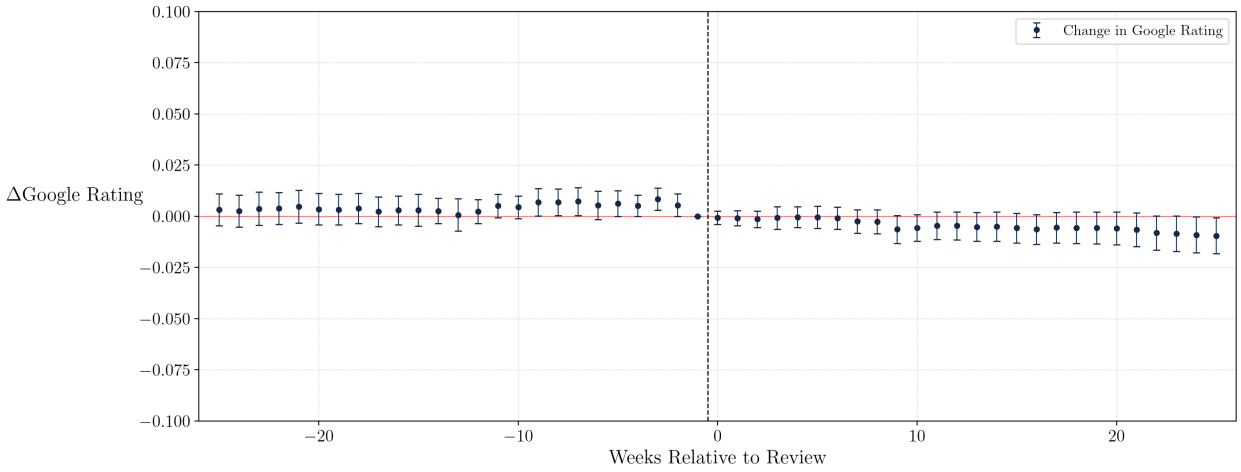



Figure A2: Average Google Rating

Notes: Figure A2 presents stacked event study estimates of the dynamic effect of a *OneBite* review on the average Google rating. The estimates show the weekly change in the displayed Google Rating in the 26 weeks before and after a review is posted, estimated on the 804 pizzerias that are matched to Li et al. (2022). Li et al. (2022) provide Google review data on establishments across the United States until August 31, 2021. The week prior to the review is normalized to 0. The confidence bars show the 95% confidence intervals with standard errors clustered at the pizzeria level.

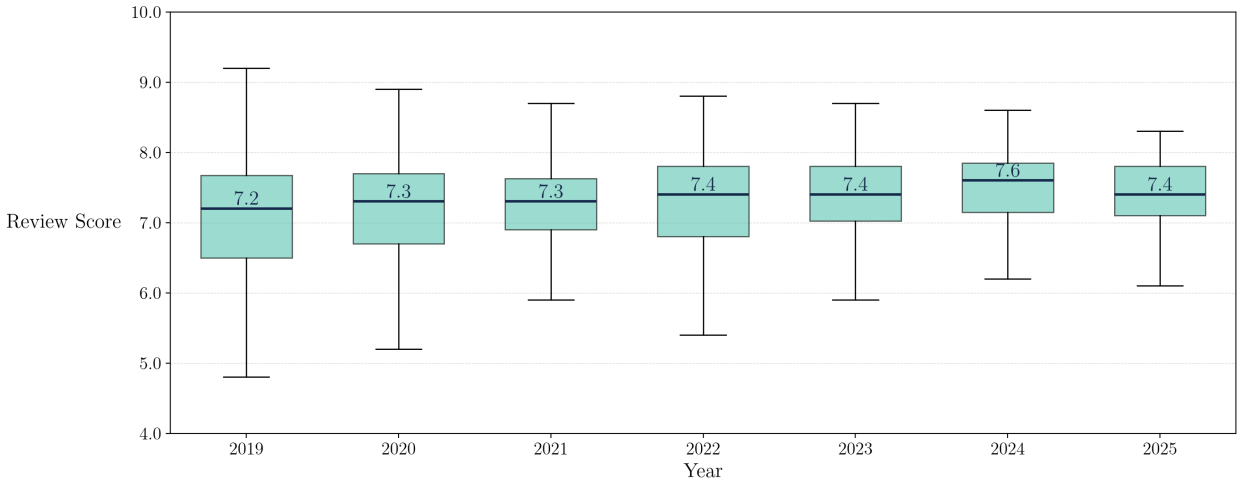
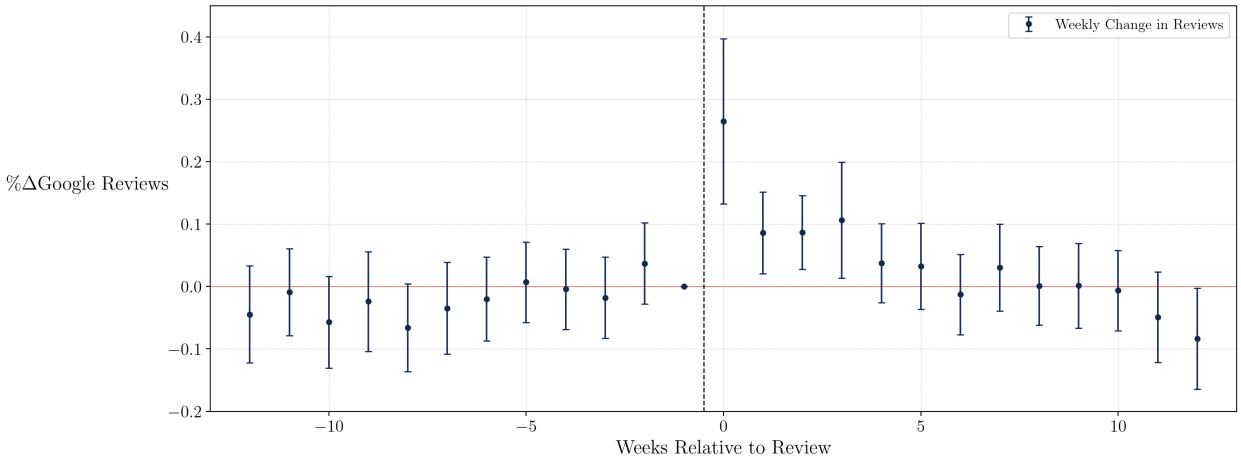
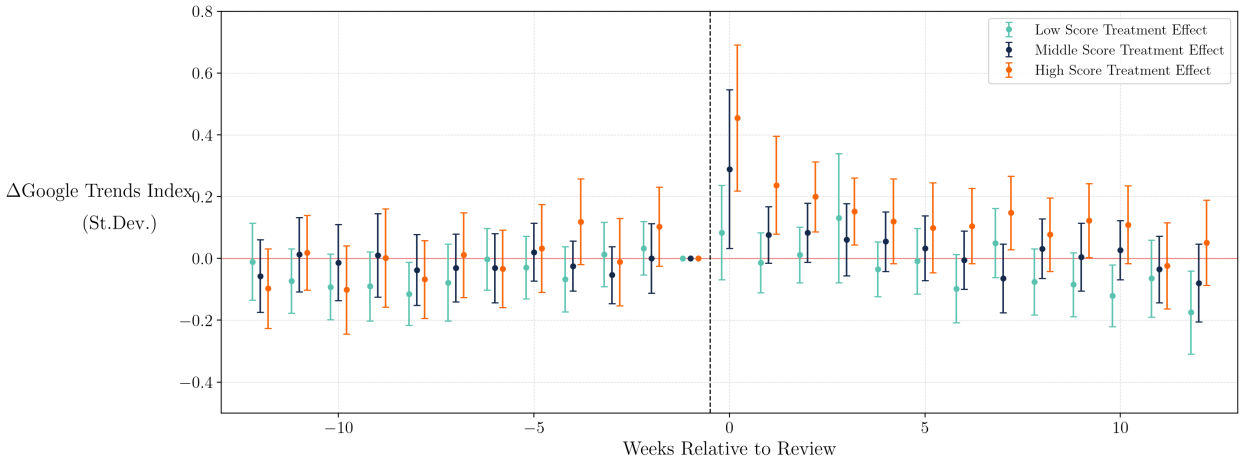



Figure A3: Score Box Plot

Notes: Figure A3 presents box plots of *OneBite* review scores by year of upload from 2019 to 2025. Each gold box shows the interquartile range of scores among reviews posted in that year, with the red line marking the median score, which is reported above each box. The whiskers extend to the minimum and maximum scores within 1.5 times the interquartile range, and outliers are omitted. Reviews with scores that cannot be determined by either the *OneBite* app or Claude's API are excluded.




(a) Any Review




(b) By Review Score

## Figure A4: Google Review Responses

Notes: Figure A4 presents stacked event study estimates of the effect of a *OneBite* review on the number of new Google reviews at a pizzeria. Panel A shows the effect of any review, while Figure A4b separates the effects by score group: $< 7.0$, $\in [7.0, 8.0)$, and $\geq 8.0$. The estimates show weekly changes over the 13 weeks before and after the review. The sample includes 804 pizzerias in the Google Trends data that I match to the Google review data from Li et al. (2022), which cover establishments across the United States through August 31, 2021. I normalize the week before the review to zero. The error bars show 95% confidence intervals, with standard errors clustered at the pizzeria level.

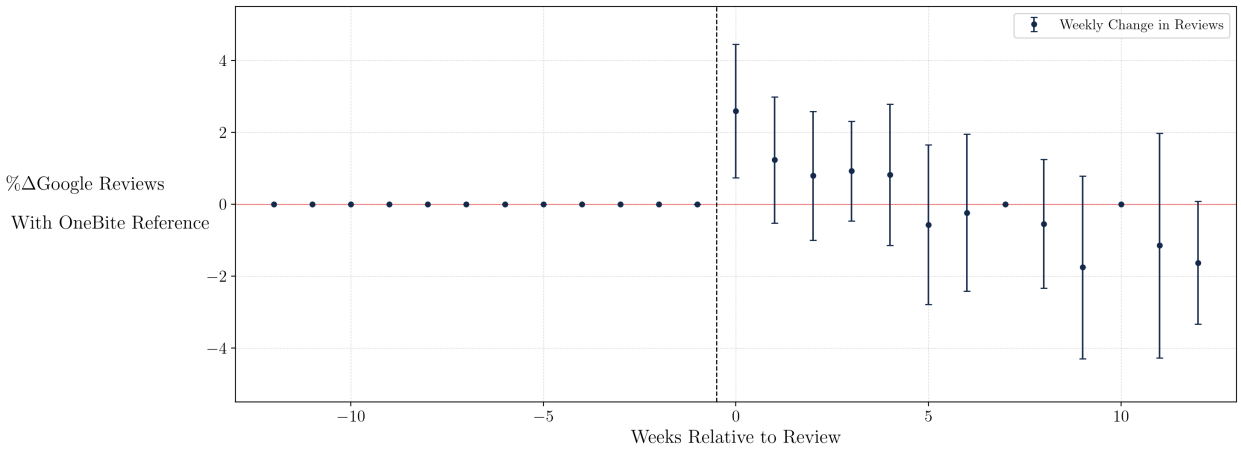



Figure A5: Google Reviews with *OneBite* Reference

Notes: Figure A5 presents stacked event study estimates of the dynamic effect of a *OneBite* review on the total number of new Google reviews at a pizzeria that include terms related to a *OneBite* review, including "Barstool Sports", "Portnoy", and "OneBite". The estimates show the weekly change in the number of reviews in the 13 weeks before and after a review is posted, estimated on the 804 pizzerias in the Google Trends data that are matched to Li et al. (2022). Li et al. (2022) provide Google review data on establishments across the United States until August 31, 2021. The week prior to the review is normalized to 0. The confidence bars show the 95% confidence intervals with standard errors clustered at the pizzeria level.

Table A1: *OneBite* Pizza Review State Counts

| State | Total Reviews | Reviews 2019-2025 |
|---|---|---|
| New York | 666 | 354 |
| New Jersey | 221 | 209 |
| Florida | 194 | 192 |
| Massachusetts | 161 | 124 |
| Illinois | 97 | 89 |
| Pennsylvania | 91 | 90 |
| Connecticut | 64 | 57 |
| California | 36 | 28 |
| Michigan | 33 | 29 |
| Ohio | 33 | 22 |
| Texas | 28 | 23 |
| Arizona | 26 | 26 |
| Nevada | 25 | 16 |
| Tennessee | 22 | 22 |
| Louisiana | 17 | 17 |
| Rhode Island | 17 | 17 |
| Minnesota | 17 | 5 |
| Kentucky | 16 | 9 |
| Wisconsin | 16 | 16 |
| Georgia | 14 | 13 |

Notes: Table A1 reports the number of *OneBite* reviews by state for the 20 states with more than 15 reviews in total or more than 10 reviews since 2019, recorded across the *OneBite* pizza app and YouTube. The first column reports the total number of reviews posted between 2013 and August 2025, while the second column reports the number of reviews posted between January 2019 and August 2025, the period covered by the SafeGraph Spend Patterns data. Minnesota is excluded from the final sample because it contains no treated pizzerias that are matched to SafeGraph's Spend Patterns dataset.

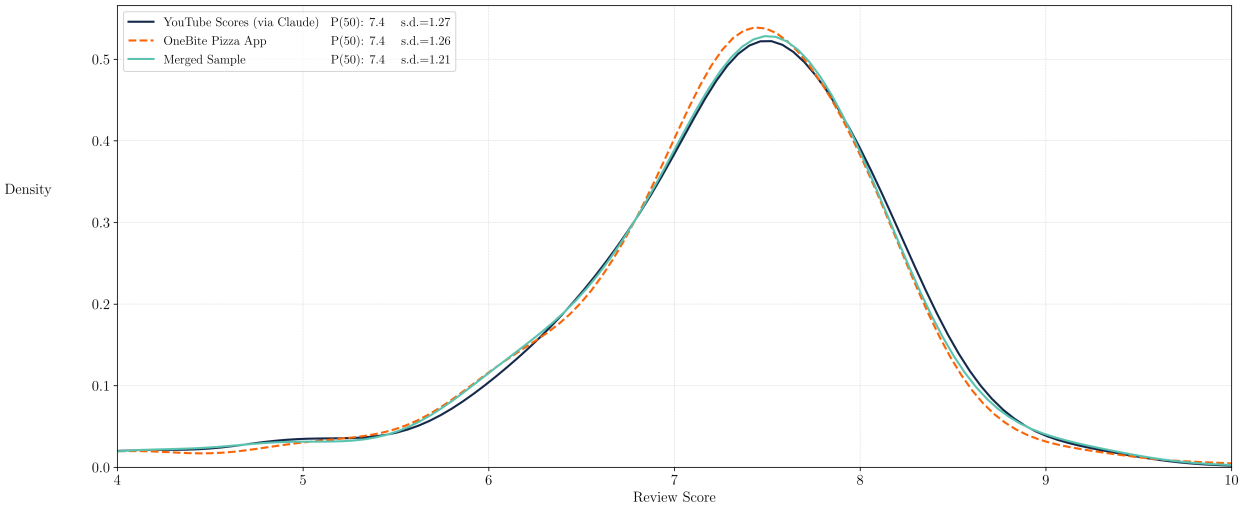
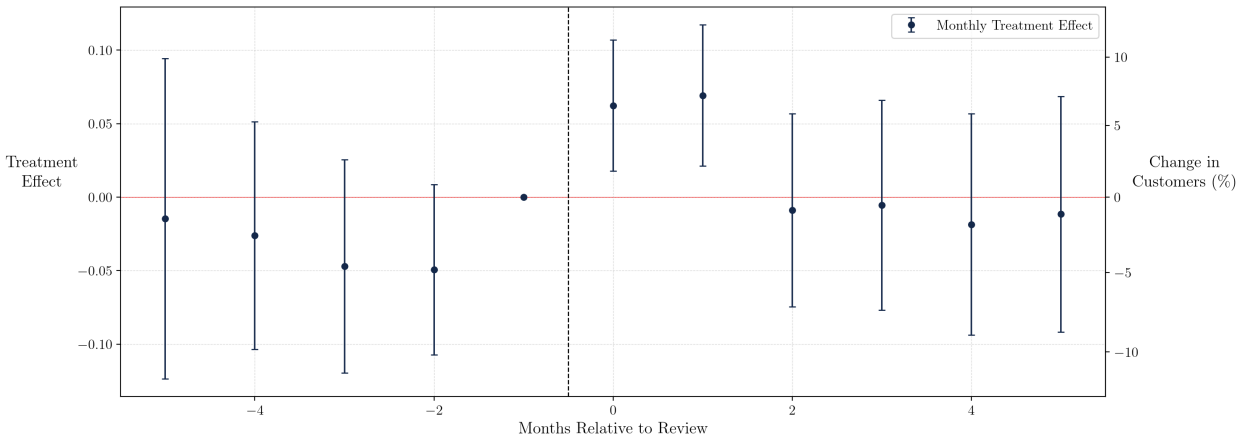
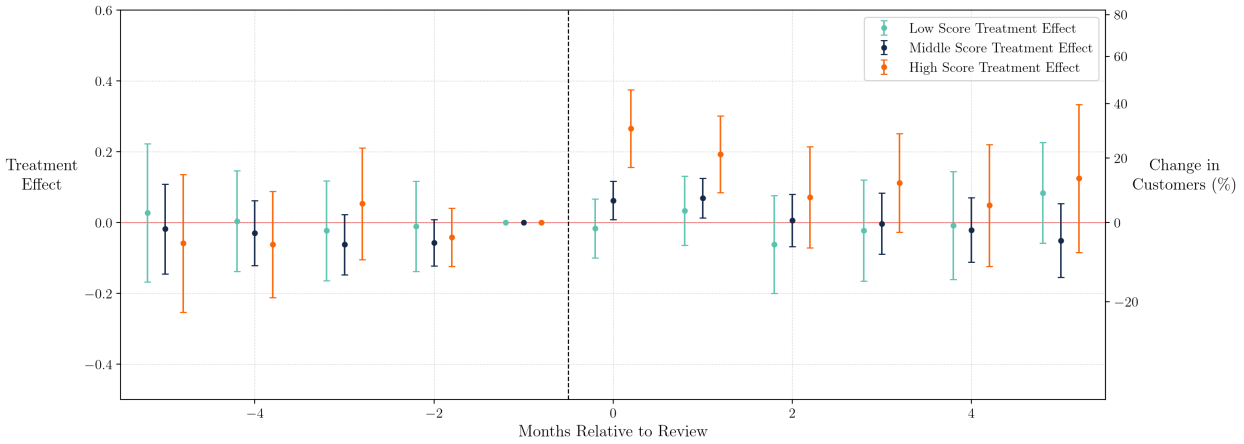



Figure A6: Density of *OneBite* Reviews

Notes: Figure A6 presents kernel density estimates of Portnoy's review scores for three samples: (1) all video reviews posted to YouTube between 2013 and 2025 across the United States, (2) all review scores posted to the *OneBite* pizza app, and (3) the review scores of pizzerias matched to the SafeGraph Spend Patterns database in the state-restricted sample.




(a) Effects of Any Review




(b) Heterogeneous Effects by Review Score

## Figure A7: Treatment Effects Using Not-Yet Treated

Notes: Figure A7 presents the event study estimates from Equation 5 to show the dynamic treatment effects of receiving a *OneBite* review, using only not-yet-treated pizzerias as the control group. Panel A shows the effects on total monthly customers in each month following treatment, whereas Panel B shows the heterogeneous treatment effects on total monthly customers by review score. The coefficient estimates are interpreted as the percentage change in each metric relative to the month prior to treatment. I correct for time and cohort heterogeneity using a stacked regression design. The confidence bars show the 95% confidence intervals with standard errors clustered at the establishment level. The left vertical axis shows the coefficient estimate from the regression, while the right vertical axis shows the percentage change interpretation using $\exp(\hat{\beta}) - 1$.

Table A2: Dynamic Treatment Effects of a *OneBite* review

| | Average Treatment Effect | Heterogeneous Treatment Effects | | |
|---|---|---|---|---|
| | (1) | (2) | | |
| | | Score$< 7.0$ | Score $\in [7.0, 8.0)$ | Score $\geq 8.0$ |
| Month -5 | 0.00717 | 0.106 | -0.00610 | -0.0970 |
| | (0.0305) | (0.0668) | (0.0349) | (0.0600) |
| Month -4 | -0.0351 | 0.0261 | -0.0490 | -0.0563 |
| | (0.0282) | (0.0566) | (0.0371) | (0.0695) |
| Month -3 | -0.0463 | -0.00991 | -0.0657 | -0.0657 |
| | (0.0273) | (0.0565) | (0.0362) | (0.0454) |
| Month -2 | -0.0394 | -0.0193 | -0.0535 | -0.0710 |
| | (0.0224) | (0.0435) | (0.0280) | (0.0506) |
| Month 0 | 0.0605* | 0.0553 | 0.0516 | 0.395*** |
| | (0.0244) | (0.0442) | (0.0340) | (0.0691) |
| Month 1 | 0.0295 | -0.0109 | 0.0142 | 0.387*** |
| | (0.0374) | (0.125) | (0.0338) | (0.0601) |
| Month 2 | -0.0206 | 0.00175 | -0.0447 | 0.295*** |
| | (0.0290) | (0.0577) | (0.0343) | (0.0659) |
| Month 3 | 0.00162 | 0.0832 | -0.0193 | 0.0496 |
| | (0.0290) | (0.0487) | (0.0402) | (0.128) |
| Month 4 | -0.00678 | 0.0903 | -0.0176 | -0.139 |
| | (0.0295) | (0.0585) | (0.0386) | (0.169) |
| Month 5 | -0.00298 | 0.118 | -0.0408 | 0.0368 |
| | (0.0326) | (0.0670) | (0.0398) | (0.170) |
| Fixed Effects | | | | |
| Pizzeria | ✓ | ✓ | ✓ | ✓ |
| Month | ✓ | ✓ | ✓ | ✓ |
| County $\times$ Month | ✓ | ✓ | ✓ | ✓ |
| Observations | 1,588,937 | 1,830,285 | 1,830,285 | 1,830,285 |

Notes: Significant at the [***] $p < 0.001$, [**] $p < 0.01$, [*] $p < 0.05$ level, with standard errors clustered at the establishment level reported in parentheses. Table A2 reports the event study coefficients after a *OneBite* review is posted to YouTube, estimated on the matched sample of pizzerias in 19 U.S. states. Column (1) reports the average treatment effects from receiving any *OneBite* review. Column (2) reports the heterogeneous treatment effects for review scores $< 7.0$, $\in [7.0, 8.0)$, and $\geq 8.0$, where all three score categories are jointly estimated. All columns correct for time and cohort heterogeneity using a stacked regression design and normalize $t = -1$ to 0. All fixed effects are also interacted with the cohort stack.

Table A3: Treatment Effects By Consumer Distance

| | Customer Distance | | | | |
|---|---|---|---|---|---|
| | (1) Within 5km | (2) 5–10km | (3) 10–25km | (4) 25–50km | (5) Over 50km |
| Average effect (Months 0–2) | 0.174 | 0.236** | 0.282** | 0.359* | 0.504*** |
| | (0.108) | (0.090) | (0.091) | (0.139) | (0.109) |
| Fixed Effects | | | | | |
| Pizzeria | ✓ | ✓ | ✓ | ✓ | ✓ |
| Month | ✓ | ✓ | ✓ | ✓ | ✓ |
| County × Month | ✓ | ✓ | ✓ | ✓ | ✓ |
| Customer Share in Untreated Sample | 27.36% | 16.13% | 19.41% | 9.25% | 27.85% |
| Observations | 1,459,116 | 1,459,674 | 1,724,420 | 1,670,653 | 1,797,036 |

Notes: Table A3 reports the average treatment effect of a high-score *OneBite* review on total monthly customers over the first three months following the review. Each column reports estimates for a different distance category, measured between the centres of the customer's home county and the pizzeria's county. I estimate a stacked PPML event study and average the coefficients for months 0–2, using their full covariance matrix to calculate standard errors. All fixed effects interact with the cohort stack. Standard errors cluster at the establishment level and appear in parentheses. The footer reports each distance category's share of customers in the untreated sample. Observations reflect the total number of observations in each stacked regression. Significant at the [***] $p < 0.001$, [**] $p < 0.01$, and [*] $p < 0.05$ levels.

## Table A4: Spillovers across Industries after a High-Score Review

| | NAICS Code | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | (1) Snack and Nonalcoholic Beverage Bars | (2) Drinking Places (Alcoholic Beverages) | (3) Miscellaneous Store Retailers | (4) Beer, Wine and Liquor Stores | (5) Grocery Stores | (6) Accommodation | (7) Specialty Food Stores | (8) Limited-Service Restaurants | (9) Full-Service Restaurants |
| Average effect (Months 0–2) | 0.150 | 0.068** | 0.039 | 0.010 | -0.018 | -0.035 | -0.061 | -0.070*** | -0.073 |
| | (0.086) | (0.021) | (0.022) | (0.016) | (0.009) | (0.026) | (0.041) | (0.011) | (0.036) |
| Fixed Effects | | | | | | | | | |
| Establishment | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Month | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County × Month | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 13.9M | 2.2M | 8.0M | 2.8M | 14.5M | 3.1M | 1.7M | 29.9M | 36.0M |

Notes: Table A4 reports the average treatment effect of a high-score *OneBite* review on total monthly customers at nearby establishments over the first three months following the review. Each column reports estimates for a separate NAICS category. The sample includes establishments within 1,000m of any reviewed pizzeria in SafeGraph's Spend Patterns data; reviewed pizzerias need not match to SafeGraph. I estimate a stacked PPML event study and normalize the month before treatment to zero. All fixed effects interact with the cohort stack. Parentheses contain standard errors clustered at the establishment level. Significance stars reflect $q$-values: [***] $q < 0.001$, [**] $q < 0.01$, and [*] $q < 0.05$. Observations report stacked sample counts in millions, rounded to one decimal place.

Table A5: Treatment Effects By Score Group

| | *OneBite* Score | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) [0, 6.4] | (2) [6.5, 6.7] | (3) [6.8, 6.9] | (4) [7.0, 7.2] | (5) [7.3, 7.4] | (6) [7.5, 7.7] | (7) [7.8, 7.9] | (8) [8.0, 10.0] |
| Average effect (Months 0–2) | -0.002 | 0.036 | 0.084 | 0.046 | -0.045 | 0.055 | 0.160*** | 0.359*** |
| | (0.102) | (0.059) | (0.082) | (0.060) | (0.054) | (0.074) | (0.036) | (0.041) |
| Fixed Effects | | | | | | | | |
| Pizzeria | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Month | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County × Month | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 251,998 | 63,012 | 60,033 | 372,950 | 395,260 | 409,295 | 115,165 | 288,861 |

Notes: Significant at the [***] $p < 0.001$, [**] $p < 0.01$, [*] $p < 0.05$ level, with standard errors clustered at the establishment level reported in parentheses. Table A5 reports the event study coefficients after a *OneBite* review is posted to YouTube, estimated on the matched sample of pizzerias in 19 U.S. states. Each treatment effect estimate measures the change in total customers after receiving a *OneBite* review score in a specific score bin: [0, 6.4],[6.5, 6.7], [6.8, 6.9], [7.0, 7.2], [7.3, 7.4], [7.5, 7.7], [7.8,7.9], [8.0, 10.0]. Observations reflect the total number of observations in the stacked PPML regression. All fixed effects are also interacted with the cohort stack.

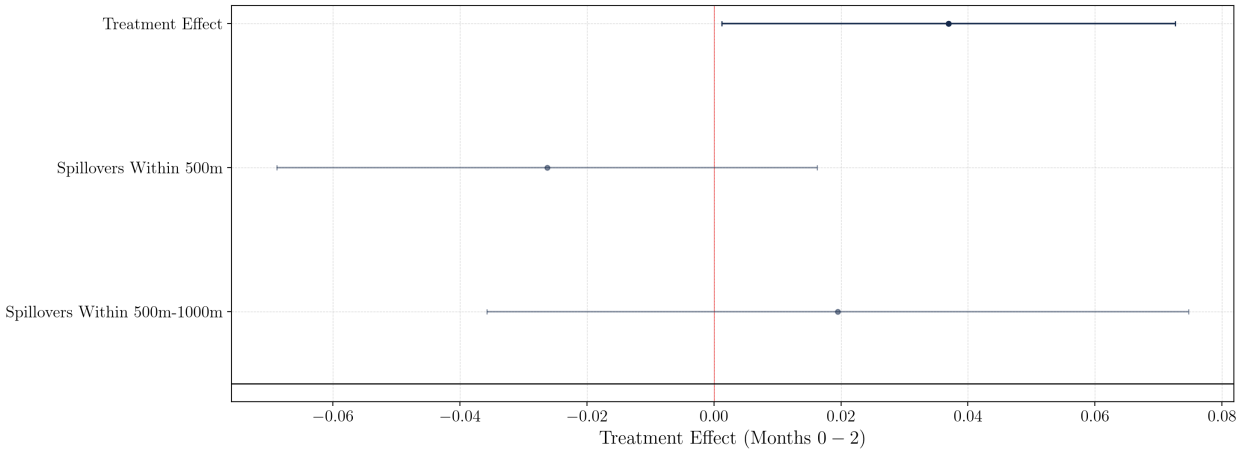



Figure A8: Limited Evidence of Substitution After Any Review

Notes: Figure A8 presents the effects on total customers over the first three months following the posting of any *OneBite* review to YouTube. The direct treatment effect on the reviewed pizzeria and spatial spillovers onto other pizzerias within 500 m and between 500 m and 1,000 m are jointly estimated following Equation 7. I correct for time and cohort heterogeneity using a stacked regression design. The confidence bars show 95% confidence intervals, with standard errors clustered at the establishment level.

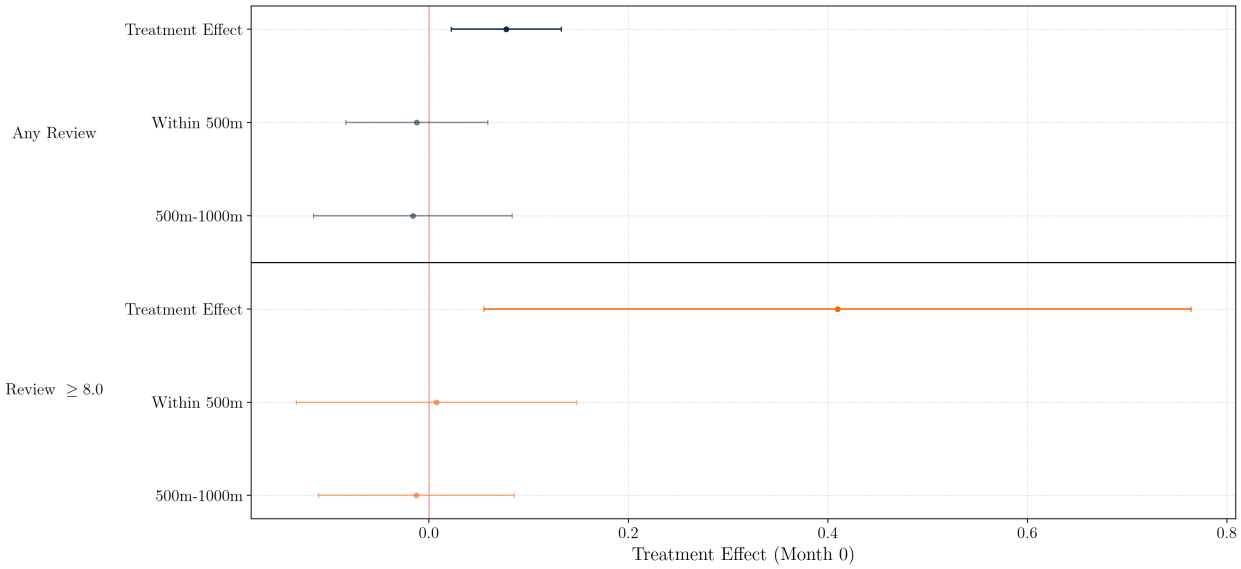



Figure A9: Limited Evidence of Substitution

Notes: Figure A9 presents the treatment effect estimates for how total customers change in the first month after a *OneBite* review is posted to YouTube, jointly estimated with spatial spillovers onto nearby establishments following Equation 7. The red bars show the treatment effect, spillovers onto pizzerias within 500m of the review, and pizzerias within a 500m to 1000m ring around the review for any review, while the gold bars present the analogous results for a review score $\geq 8.0$. I correct for time and cohort heterogeneity using a stacked regression design. The confidence bars show the 95% confidence intervals with standard errors clustered at the establishment level.

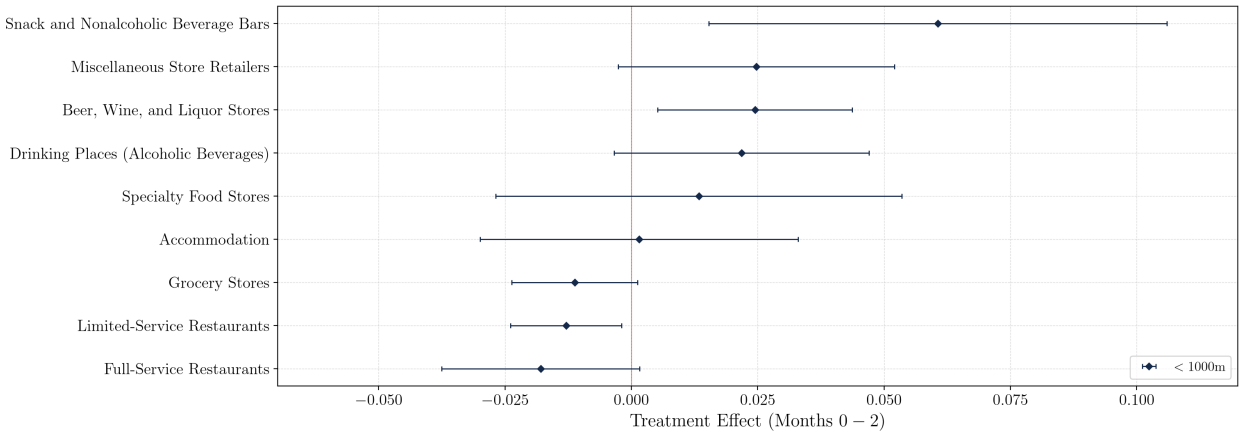


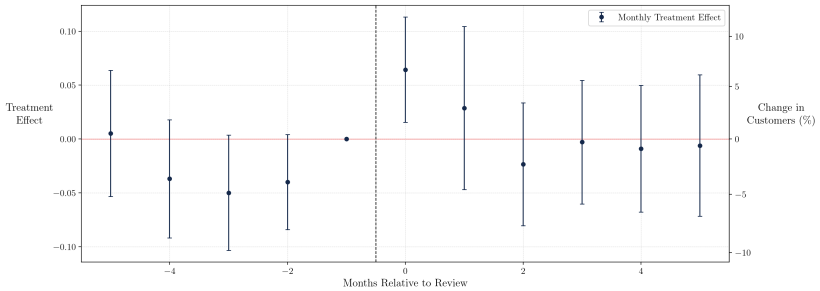

Figure A10: Spillovers across Industries after Any Review

Notes: Figure A10 presents the average treatment effect over the first three months following any *OneBite* review on total monthly customers at nearby establishments, estimated separately by the industry of the nearby establishment, where industries are defined by NAICS code. The sample includes all establishments within 1,000 m of any reviewed pizzeria in SafeGraph's Spend Patterns data, where reviewed pizzerias need not be matched. I correct for time and cohort heterogeneity using a stacked regression design and normalize $t = -1$ to 0. The confidence bars show 95% confidence intervals adjusted for multiple hypothesis testing using $q$-values, with standard errors clustered at the establishment level.

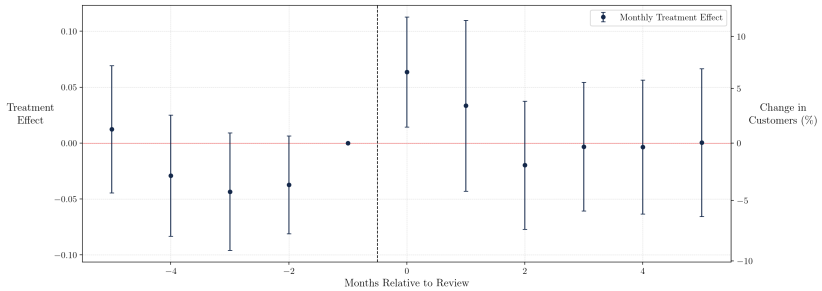
Table A6: Change in Google Reviewers

| | Share (%) | | | Total Count | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Firm | Pre-Review | Post-Review | Difference | Pre-Review | Post-Review | Difference |
| Pizza restaurant | 95.0 | 95.1 | 0.13 | 6,446 | 7,372 | +14.4% |
| Restaurant | 87.6 | 86.4 | -1.19* | 5,942 | 6,694 | +12.7% |
| Coffee shop | 54.6 | 53.3 | -1.30 | 3,701 | 4,126 | +11.5% |
| Clothing store | 39.1 | 38.2 | -0.97 | 2,654 | 2,956 | +11.4% |
| Event venue | 41.8 | 40.6 | -1.16 | 2,836 | 3,149 | +11.0% |
| Tourist attraction | 55.1 | 53.5 | -1.62* | 3,739 | 4,145 | +10.9% |
| Sporting goods store | 22.6 | 21.9 | -0.68 | 1,534 | 1,699 | +10.8% |
| Department store | 35.1 | 33.9 | -1.13 | 2,378 | 2,628 | +10.5% |
| Bar | 70.4 | 67.7 | -2.74*** | 4,776 | 5,242 | +9.8% |
| Park | 39.5 | 38.0 | -1.55* | 2,681 | 2,942 | +9.7% |
| Hotel | 42.1 | 40.0 | -2.09** | 2,859 | 3,103 | +8.5% |
| Movie theater | 26.2 | 24.5 | -1.69* | 1,778 | 1,900 | +6.9% |
| Shopping mall | 36.5 | 34.1 | -2.43** | 2,477 | 2,641 | +6.6% |
| Museum | 20.9 | 18.9 | -1.94** | 1,415 | 1,466 | +3.6% |

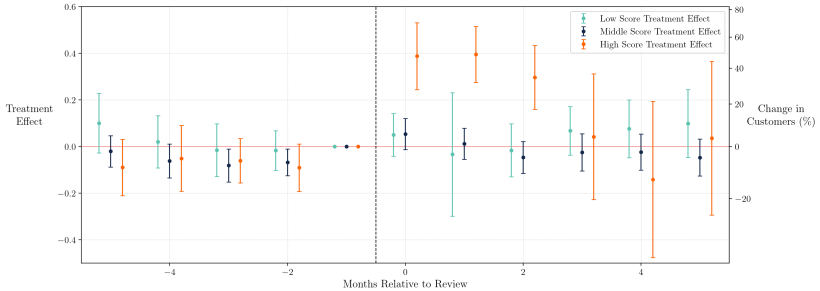
Notes: Table A6 compares individuals who posted Google reviews of treated pizzerias in the three months before and after a *OneBite* review, examining the other establishment types these individuals reviewed. Although Google distinguishes over 4,000 establishment categories, I aggregate these into 14 broad categories, combining detailed classifications such as different types of clothing stores. I retain area pizza restaurants as a separate category to compare reviewers' engagement with other pizzerias. Columns (1) and (2) report the share of reviewers in each group who reviewed each establishment type. Column (3) reports the difference in shares, with statistical significance based on multiple-testing-adjusted $q$-values. Significance levels are denoted by *** $q < 0.01$, ** $q < 0.05$, and * $q < 0.10$. Columns (4) and (5) report the corresponding reviewer counts, and Column (6) reports their percentage growth from the pre-review to the post-review group.



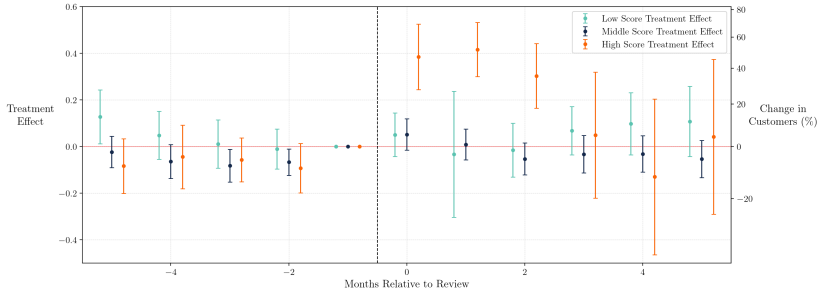

(a) Excluding Pizzerias within 500m (Any Review) Review




(b) Excluding Pizzerias within 1,000m (Any Review)



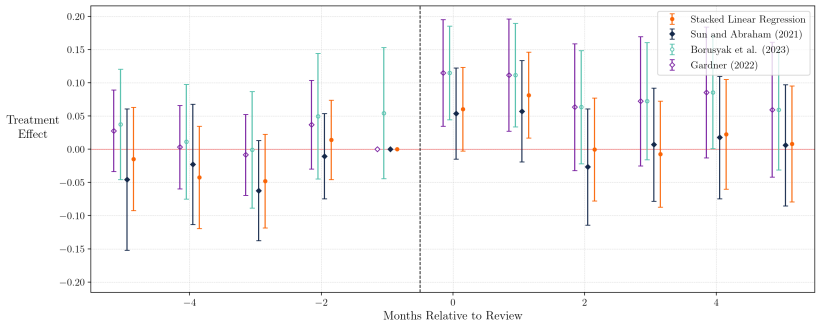

(c) Excluding Pizzerias within 500m (By Score)



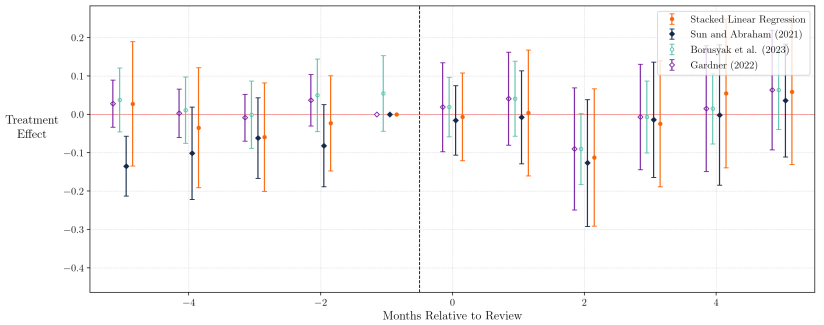

(d) Excluding Pizzerias within 1,000m (By Score)

Figure A11: Excluding Nearby Control Pizzerias

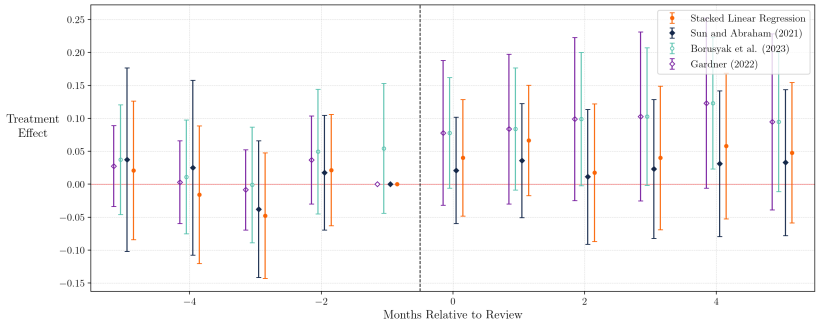
Notes: Figure A11 presents the event study estimates from Equation 5 to show the dynamic treatment effects of receiving a *OneBite* review on total monthly customers in each month following treatment, where the coefficients are interpreted as the percentage change in total monthly customers relative to the month prior to treatment. Figure A11a excludes untreated pizzerias within 500m of a treated pizzeria from the control group, while Figure A11b excludes untreated pizzerias within 1,000m. Figures A11c and A11d present the analogous event studies for the heterogeneous treatment effects specification. I correct for time and cohort heterogeneity using a stacked regression design and normalize $t = -1$ to 0. The confidence bars show the 95% confidence intervals with standard errors clustered at the establishment level.



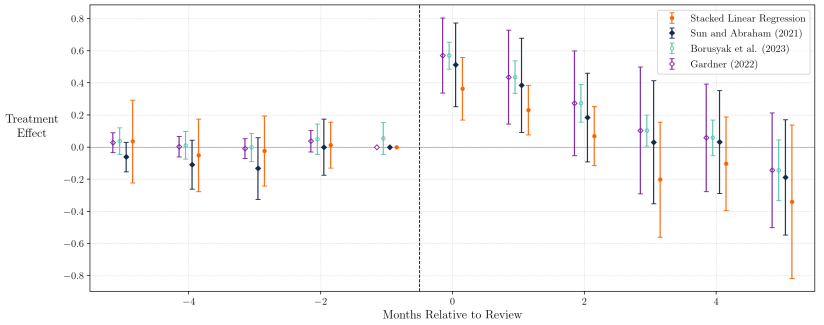

(a) Any *OneBite* Review




(b) Score $< 7.0$




(c) Score $\in [7.0, 8.0)$




(d) Score $\geq 8.0$

Figure A12: Additional Staggered Event Study Corrections

Notes: Figure A12 presents the alternative event study estimates using different strategies to correct for time and cohort heterogeneity. While the main specification follows a PPML specification, most other corrections require a linear specification; hence, the dependent variable in all equations is $\ln$ Customers. The 4 panels present the results for: Any Review, a Low-Score Review, a Middle-Score Review, and a High-Score Review. The red hollow diamonds use the correction following Gardner (2022), the green hollow circles use the correction following Borusyak et al. (2023), the dark maroon solid diamonds use the correction following Sun and Abraham (2021), and the gold solid circle estimates use a stacked regression with a linear specification. The confidence bars show the 95% confidence intervals with standard errors clustered at the establishment level.

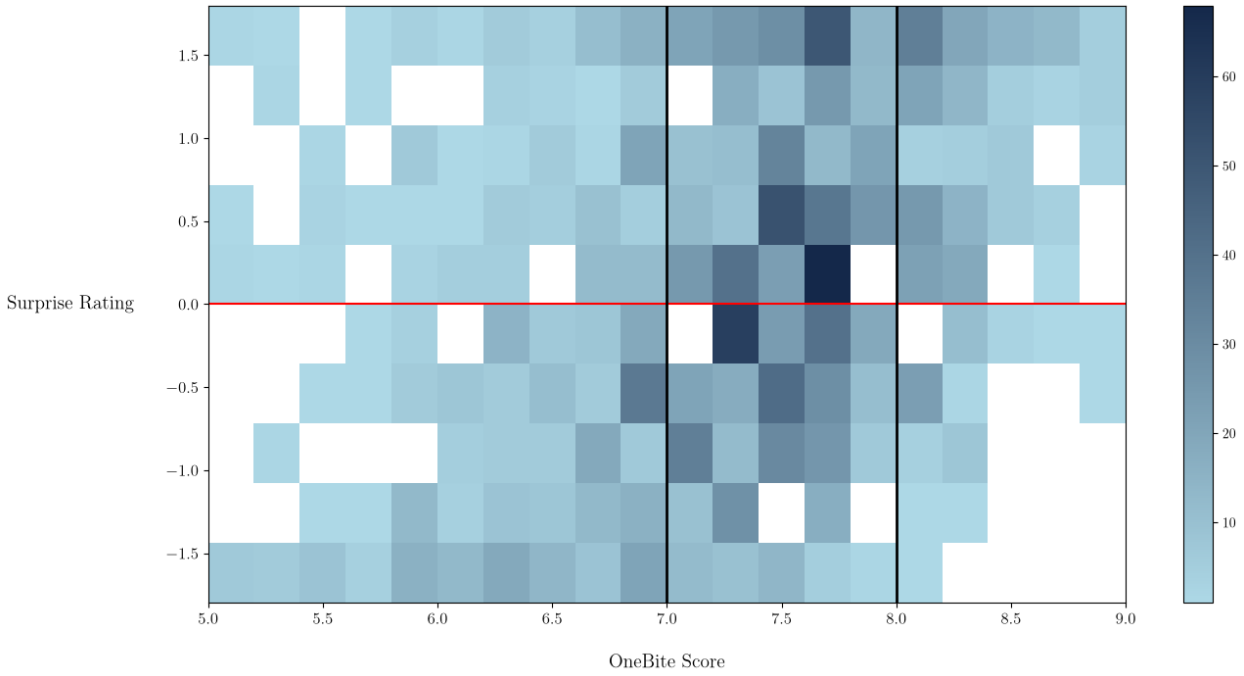



Figure A13: Information Update from a *OneBite* Review

Notes: This heat map plots the information surprise against the *OneBite* score. The vertical axis measures surprise as the *OneBite* score minus the rescaled Google rating, in *OneBite* score units. I rescale Google ratings to match the mean and standard deviation of *OneBite* scores. Red lines separate positive and negative information updates, while black lines mark the review score categories. Figure A14 presents a complementary heatmap that shows how *OneBite* scores vary with the Google Rating.

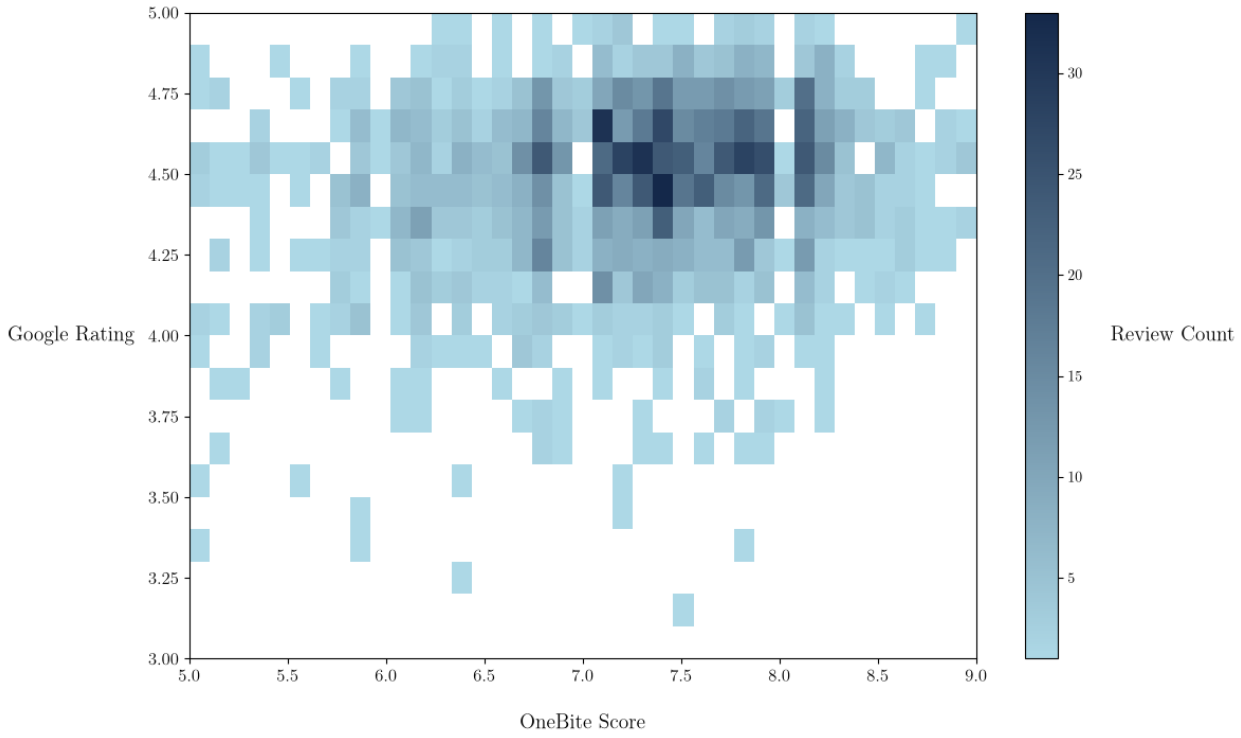
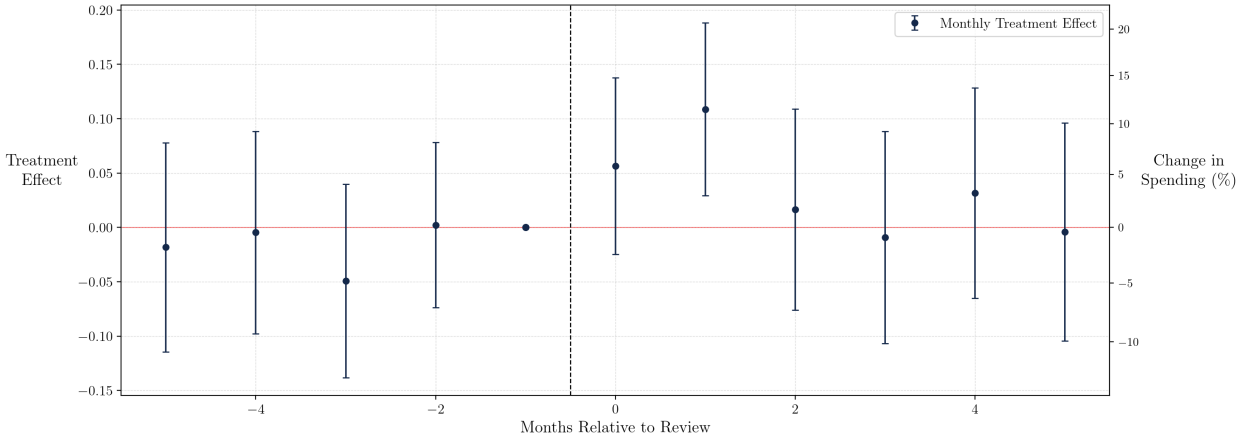
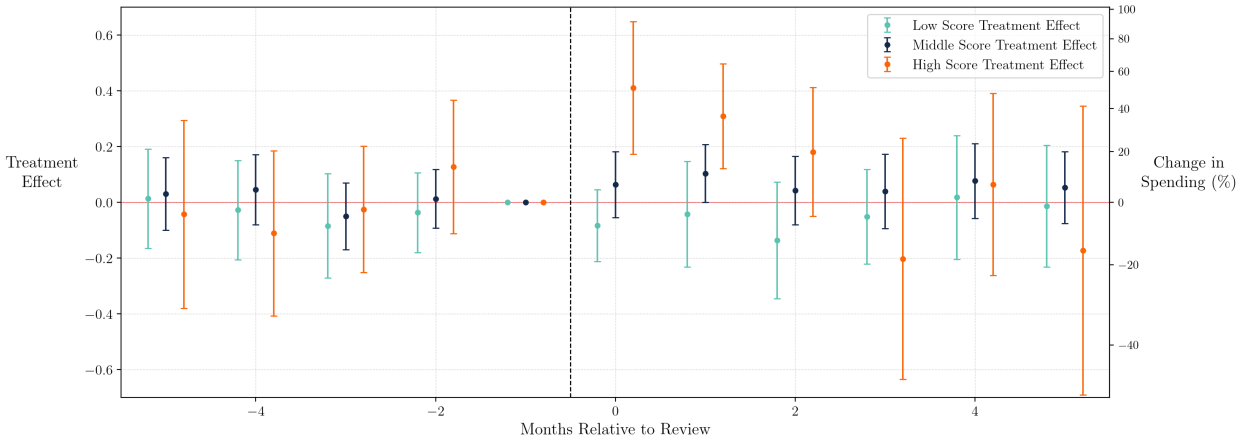



Figure A14: Google Ratings Heatmap

Notes: Figure A14 presents a heatmap to show how each pizzeria's online Google rating compares to Portnoy's review score. The vertical axis shows the Google rating in 0.1 increments, while the horizontal axis shows Portnoy's score in 0.1 increments. The underlying data uses all pizzerias that receive a *OneBite* review, where the score is either recorded on the app or imputed from the YouTube transcript. I further restrict the sample to include the establishments with a score extracted by Claude, if they are classified as a restaurant in the Google Places API.




(a) Any Review




(b) Heterogeneous Treatment Effects By Score

## Figure A15: Effects on Total Spending

Notes: Figure A15 presents the event study estimates to show the dynamic treatment effects of receiving different *OneBite* review scores. Panel A shows the effects on total monthly spending for any review in each month following treatment, whereas Panel B shows the effect on total monthly spending based on the review score. I correct for time and cohort heterogeneity using a stacked regression design. The confidence bars show the 95% confidence intervals with standard errors clustered at the establishment level. The left vertical axis shows the coefficient estimate from the regression, while the right vertical axis shows the percentage change.

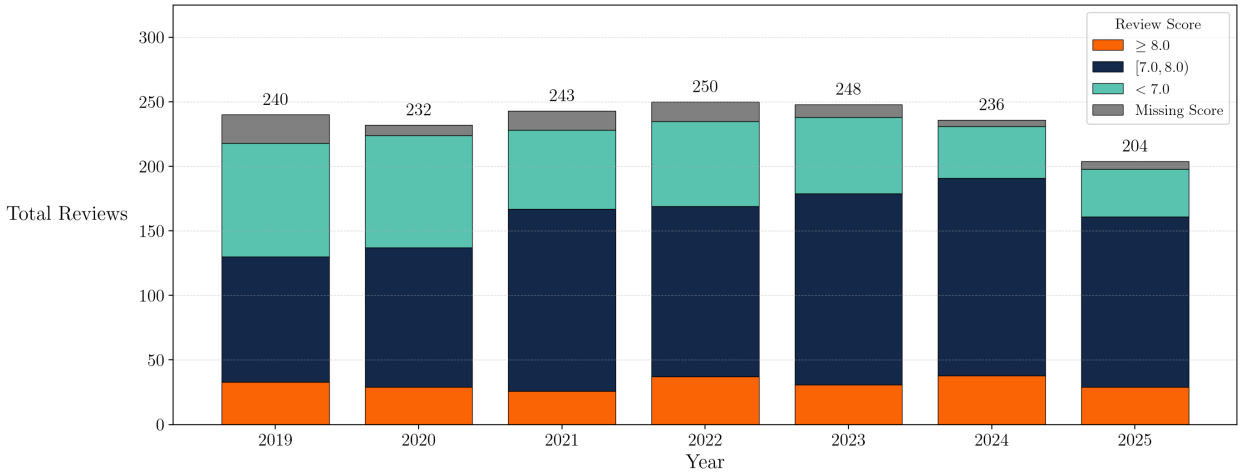



Figure A16: *OneBite* Reviews by Year

Notes: Figure A16 presents the number of *OneBite* reviews posted in each year from 2019 to 2025, where each bar is decomposed into the three score categories $< 7.0$, $\in [7.0, 8.0)$, and $\geq 8.0$. The total number of reviews in each year is reported above each bar. The grey portion of each bar shows reviews whose scores cannot be determined from either the *OneBite* app or Claude's API.

Table A7: Spatial Spillovers onto Nearby Pizzerias

| | *OneBite* Review Treatment | |
|---|---|---|
| | (1) | (2) |
| | Any Review | Review score $\geq 8.0$ |
| Treatment Effect | 0.037 | 0.370** |
| | (0.025) | (0.141) |
| Spillover: 500m | -0.026 | -0.015 |
| | (0.034) | (0.063) |
| Spillover: 1,000m | 0.019 | 0.003 |
| | (0.044) | (0.058) |
| Fixed Effects | | |
| Pizzeria | ✓ | ✓ |
| Month | ✓ | ✓ |
| County × Month | ✓ | ✓ |
| Observations | 3,244,903 | 1,115,278 |

Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Standard errors clustered at the establishment level are reported in parentheses. The table reports average effects over event months 0–2 following the posting of a *OneBite* review to YouTube, estimated on the matched sample of pizzerias in 19 U.S. states. Column (1) reports treatment and spillover effects from any review, while column (2) reports the corresponding effects from a high-score review (score $\geq 8.0$). Each specification jointly estimates the direct effect on the reviewed pizzeria and spillover effects on other pizzerias within 500 m and between 500 m and 1,000 m. Observations reflect the total number of observations in each stacked PPML regression. All fixed effects are interacted with the cohort stack.

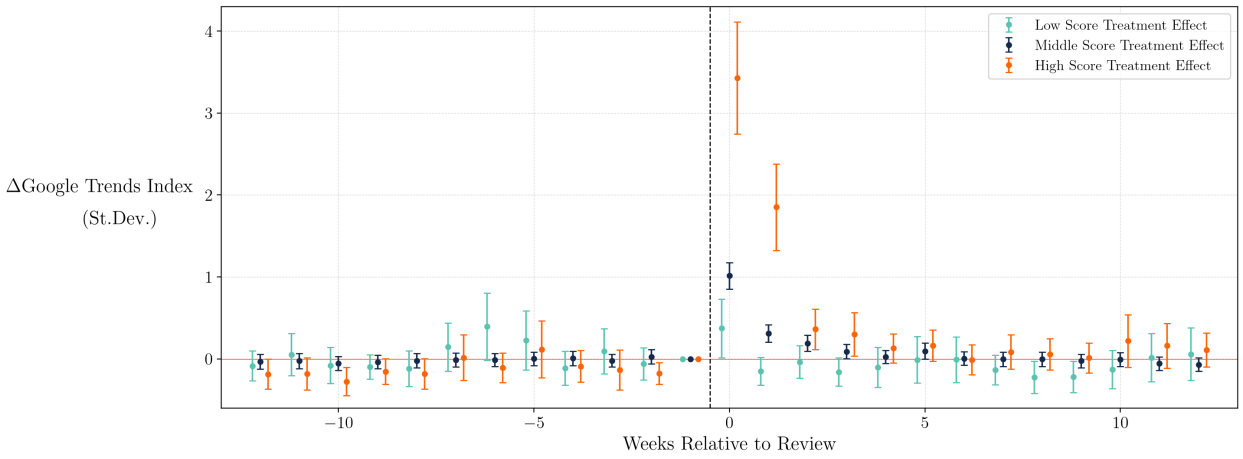



Figure A17: Google Trends Response by Score Category

Notes: Figure A17 presents stacked event study estimates of the dynamic effect of a *OneBite* review on local search attention, measured using Google Trends data within a designated market area. The figure splits the change in trends across Portnoy's three score categories: $< 7.0$, $\in [7.0, 8.0)$, and $\geq 8.0$. The estimates show the weekly change in Google Trends search index in standard deviations in the 13 weeks before and after a review is posted, estimated on the geographic Google Trends from 1263 reviews across 95 Designated Media Markets (DMAs). Reviews are excluded from the sample when the pizzeria's location cannot be assigned to a DMA or when Google Trends returns no DMA-level search data for the query window. The search index ranges from 0 to 100 across DMAs within each time-window query. The week prior to the review is normalized to 0. The confidence bars show the 95% confidence intervals with standard errors clustered at the DMA level.

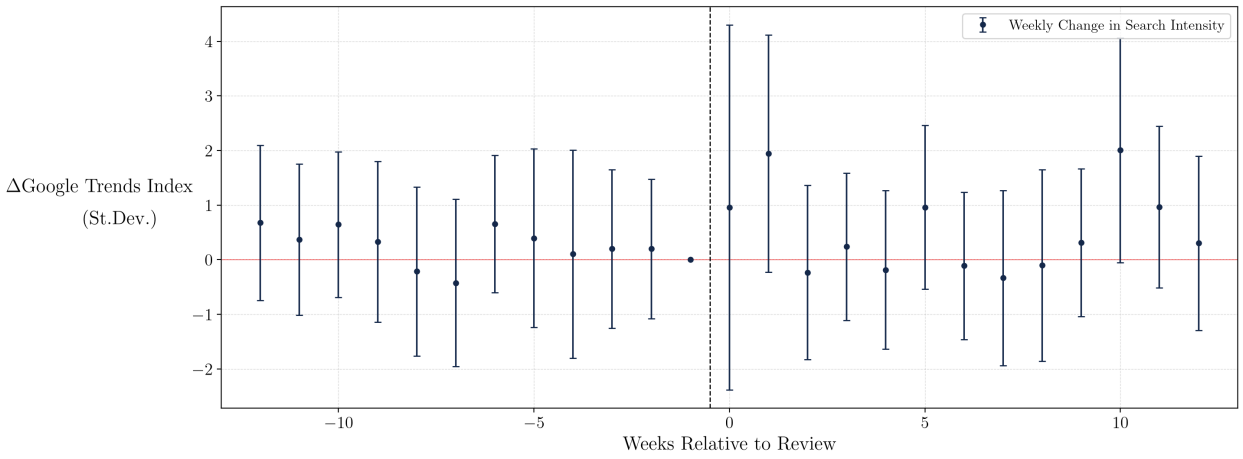


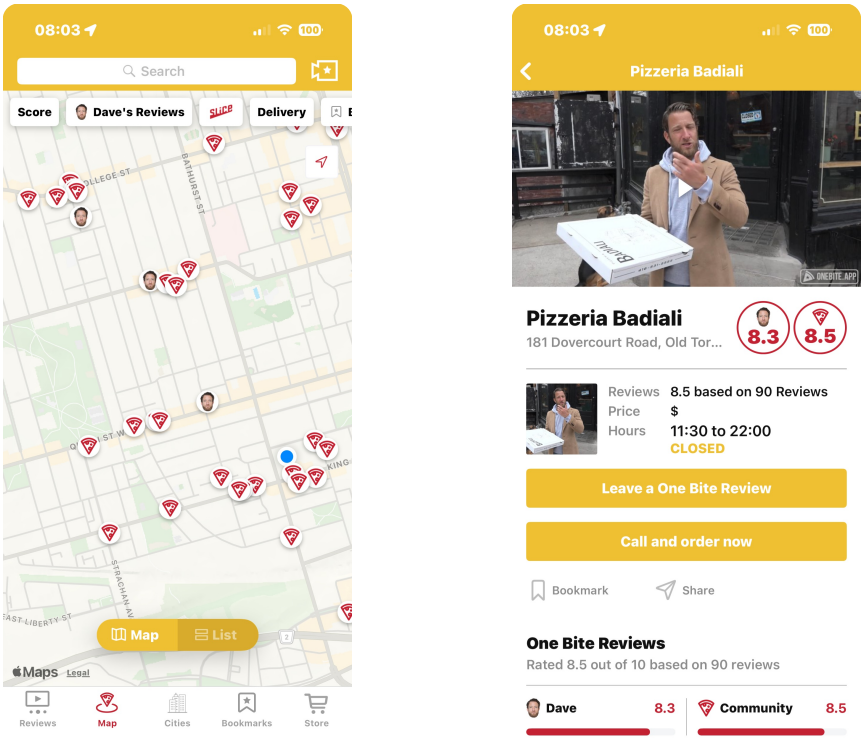

Figure A18: Google Trends Difference-in-Discontinuities Around 8.0

Notes: Figure A18 presents stacked event study estimates from a difference-in-discontinuities regression to evaluate the change in Google Trends around the 8.0 high-score cut-off. The sample only includes pizzerias between 7.7 and 8.2, and allows for different linear slopes on either side of the cut-off. The estimates show the weekly change in Google Trends search index in standard deviations in the 13 weeks before and after a review is posted, estimated on the geographic Google Trends from 142 reviews in the score range across 33 Designated Media Markets (DMAs). Reviews are excluded from the sample when the pizzeria's location cannot be assigned to a DMA or when Google Trends returns no DMA-level search data for the query window. The search index ranges from 0 to 100 across DMAs within each time-window query. The week prior to the review is normalized to 0. The confidence bars show the 95% confidence intervals with standard errors clustered at the DMA level.

# B Data Appendix

## B.1 *OneBite* App Data and Transcript Scores

In Appendix B.1, I describe how I collect data from the *OneBite* app and YouTube transcripts. Figure B1 presents the *OneBite* app's interface to show the information available for all users. The map in Figure B1a shows all reviewed pizzerias in the user's area, with Dave Portnoy icons to mark a pizzeria with a *OneBite* review, and the pizza slice icons show app-user-rated pizzerias without a *OneBite* review. Figure B1b shows an example of an information page for a pizzeria: Pizzeria Badiali. Each page lists Portnoy's review score, the average app-user score, the pizzeria's address and hours of operation, and a link to the video review. I web-scrape the data for all pizzerias listed on the app, where the webpage provides the pizzeria's review details and the map provides the pizzeria's exact geographic coordinates.




(a) Pizzeria Map

(b) Pizzeria Page

Figure B1: *OneBite* App

Notes: Figure B1 presents screenshots of the *OneBite* app showing Pizzeria Badiali in Toronto, Ontario. Panel B1a displays the map interface; Portnoy icons identify pizzerias he reviewed, while pizza-slice icons identify pizzerias reviewed only by app users. Panel B1b displays Pizzeria Badiali's information page, including review scores, location details, and the video-review link.

Although the *OneBite* app includes all review details, the app omits some pizzerias with video reviews on YouTube. To complete the treatment sample, I web-scrape all video titles, views, upload dates, and transcripts from the *OneBite* YouTube channel, excluding any YouTube Shorts and non-pizzeria reviews. Using the web-scraped data, I extract each pizzeria's name and city from the video's title, which typically follows the format "Barstool Pizza Review - Pizzeria Badiali (Toronto, ON)." I then match the video reviews to the *OneBite* app scores using the name, video's upload date, and pizzeria's city. For any unmatched video reviews, I manually extract the data from the video's infographic shown in Figure B2, which includes the pizzeria's address. I then query the Google Places API using the pizzeria's name and address to obtain its geographic coordinates. Finally, I merge the augmented video data with the app records to complete the treatment sample.




Figure B2: *OneBite* Review Infographic

Notes: Figure B2 presents the video infographic from the *OneBite* review of Pizzeria Badiali in Toronto, Ontario. The infographic contains specific geographic data for the reviewed pizzeria, including its address, city, and state/province.

After the final match, 791 unmatched *OneBite* reviews remain on YouTube that are not on the app. To impute the score, I use Claude's Haiku 4.5 to read the video's transcript and extract the awarded score using common phrases Portnoy uses in his videos. Alongside the transcript, I provide Claude with the prompt below:

> You are extracting pizza review scores from auto-generated YouTube transcripts. Dave Portnoy reviews pizzerias and announces a score from 0.0 to 10.0, typically at the end. It sometimes comes after 'I'm gonna go', or before 'that's a review'. The transcript is ASR-generated and may be garbled. Find the score he gives and return ONLY a number like 7.4 or 8.1, but the text may not be a number. If you cannot find a score, return null. No explanation.

To validate Claude's accuracy, I extract review scores from all YouTube transcripts and compare the extracted scores to the app scores. Figure B3 presents a heatmap of the scores, where each cell represents a 0.1 change in score. Although some deviation exists, Claude's extracted score from the transcript is within 0.1 points of the known app score for over 93% of matched pizzerias. With the extracted scores, I complete the treatment

sample with all treated pizzeria review locations, upload dates, and scores.

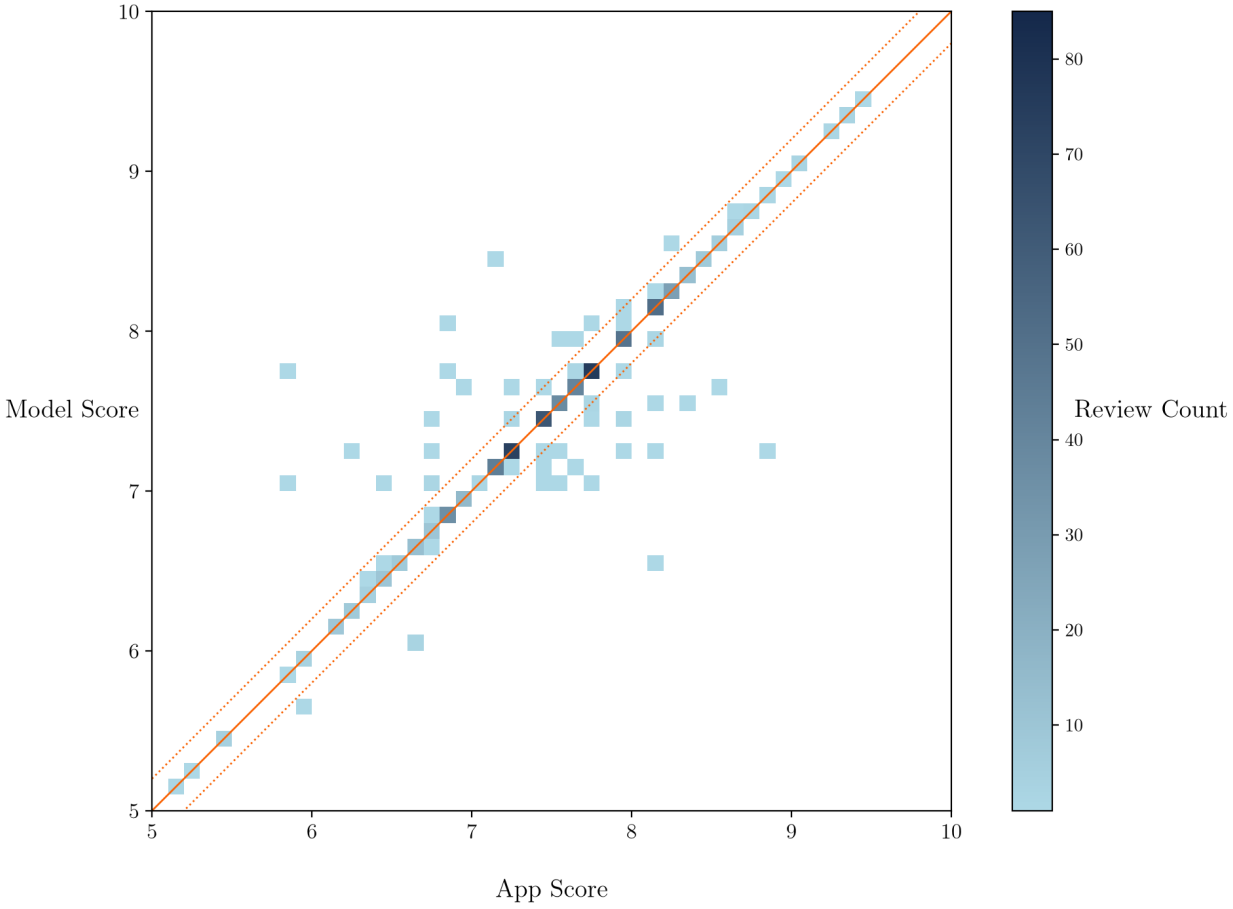



Figure B3: Claude Score Extractions

Notes: Figure B3 presents a heatmap to show Claude's extracted score from a video's YouTube transcript compared to the app score. The prompt instructs the model to identify Portnoy's score from common surrounding phrases, such as "I'm gonna go [score]" or "[score], that's a review," within otherwise unstructured text. Each cell represents a 0.1 change in score. Cells that lie directly on the solid 45-degree line indicate scores that Claude perfectly predicts, while cells that lie within the dotted lines show extracted scores that are within 0.1 of the app score.

## B.2 SafeGraph Spend Patterns Data

Table B1 presents summary statistics for SafeGraph's Spend Patterns data and highlights two features. First, the monthly customer counts, spending, and transactions reflect activity from an underlying panel of approximately 4% of U.S. consumers, rather than total activity at each establishment. As such, these measures capture only a fraction of each establishment's activity. Second, the distributions are skewed to the right, with large observations contributing to the high standard deviations. The robustness checks using log

Table B1: Summary Statistics of SafeGraph Spend Patterns

| | | | Percentile | | | | |
|---|---|---|---|---|---|---|---|
| Variable | Mean | Standard Deviation | $10^{th}$ | $25^{th}$ | $50^{th}$ | $75^{th}$ | $90^{th}$ |
| Customers | 67.2 | 96.2 | 4.3 | 10.9 | 31.8 | 80.7 | 168.8 |
| Spending ($) | 2,845.0 | 4,190.2 | 157.0 | 429.4 | 1,292.2 | 3,357.5 | 7,224.1 |
| Transactions | 90.6 | 132.9 | 5.2 | 13.9 | 41.5 | 107.8 | 229.0 |
| Spending Per Transaction ($) | 35.2 | 24.5 | 17.1 | 23.1 | 30.0 | 40.1 | 57.0 |

Notes: Table B1 reports summary statistics for 295,069 pizzeria-month observations from SafeGraph's Spend Patterns dataset in my final sample. For this table, I winsorize customers, spending, and transactions at the 99th percentile.

customers as the dependent variable compress the right tail and yield statistically similar results, which suggests that the skewed distributions do not pose a concern for the main results.

While SafeGraph's Spend Patterns dataset provides rich data on consumer spending, one limitation that may confound the treatment effects is that privacy agreements with data providers suppress some establishment-month records. While the missing observation does not necessarily indicate zero economic activity or firm closure, the suppression creates an unbalanced panel. If observations are missing nonrandomly, the estimated treatment effects may reflect changes in the composition of the observed treated cohort rather than a causal effect of the review. To determine whether a *OneBite* review changes which pizzerias appear in the observed sample, I estimate three staggered event studies for the probability of appearing in the sample, transitioning from missing to positive customer counts, and transitioning from positive to missing customer counts. Although the aggregate effect is the parameter of interest, I separately investigate positive-to-missing transitions and missing-to-positive transitions to ensure that sample entry and exit do not offset one another in the aggregated effect. Table B2 reports the estimates from the stacked event studies over the first two treatment months, as video uploads occur at any point within a month.

The results in Table B2 provide evidence that sample entry and exit do not correspond to the review date. First, the net extensive margin reports the effect of a *OneBite* review on the probability that the pizzeria appears in the sample. The point estimate averaged over the first two months following treatment is close to zero and statistically insignificant, with no evidence that a review changes the probability that a pizzeria reports customers in the sample. While statistical insignificance alone does not establish the absence of an effect, the narrow confidence intervals rule out economically meaningful

Table B2: No Evidence of Non-Random Sample Entry/Exit

| Extensive Margin Test | Estimate | Std. Error | 95% Confidence Interval |
|---|---|---|---|
| *Stacked Event-Study* [*Months 0 and 1*] | | | |
| Net Extensive Margin | 0.0004 | (0.0106) | [-0.0203, 0.0211] |
| Missing-to-Positive | -0.0075 | (0.0099) | [-0.0269, 0.0119] |
| Positive-to-Missing | -0.0040 | (0.0090) | [-0.0217, 0.0137] |

Notes: Table B2 presents the results from several statistical tests to evaluate whether a *OneBite* review affects the extensive margin. The estimates show results from a staggered event study on the extensive margin of customers, correcting for time and cohort heterogeneity using a stacked regression specification. The three rows report the net extensive-margin effect and the transition probabilities from missing to positive customers and from positive to missing customers in the first two months following a review. The final column presents the 95% confidence interval with standard errors clustered at the pizzeria level.

changes Romer (2020). The 95% confidence interval implies that a review changes the probability of exiting the observed sample by no more than 2.2 percentage points relative to the control group. I then investigate positive-to-missing transitions separately from missing-to-positive transitions to ensure that sample entry and exit do not offset one another in the aggregated effect. Similar to the net extensive margin, both point estimates are close to zero and statistically insignificant. Furthermore, the narrow confidence interval implies that a review does not materially change the likelihood that a pizzeria transitions out of or into the sample. While data suppression may cause pizzerias to exit the sample, the timing is not correlated with a *OneBite* review.